
AV Fundamentals for Educators

Practice Exam and Assessment Bank

ISTE International Society for Technology in Education × **AVIXA** Audiovisual and Integrated Experience Association

120-Question Assessment Package | Aligned to the AVIXA AV Technologist Test Domain Blueprint | Study Aid — Not an Official AVIXA Exam

Prepared: May 2026 | For educational and professional-development use only

Disclaimer

This document is an **independent study aid** aligned to the publicly available AVIXA AV Technologist Test domain blueprint. It is *not* produced, endorsed, or approved by AVIXA as an official exam. Question content draws on industry-standard AV knowledge consistent with the domains tested. ISTE branding reflects the pedagogical framework and educational context of this resource. All trademarks are property of their respective owners.

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Front Matter: Instructions and Domain Blueprint

Instructions for Exam Takers

This practice exam is designed to help you assess your readiness for the **AVIXA AV Technologist Test** and to build mastery of the core audiovisual concepts covered in the *AV Fundamentals for Educators* curriculum. The package contains two components:

- **100-Question Practice Exam** — a full-length, timed exam simulation covering all nine AVIXA domains. Allow **90 minutes** of uninterrupted time.
- **20-Question Quick-Check Bank** — five short quizzes (4 questions each), one per course unit, for formative self-assessment after each unit of study.

Before You Begin — Exam Conditions

1. Find a quiet location free from distractions.
2. Set a 90-minute timer for the 100-question exam.
3. Read each question stem carefully before reviewing the options.
4. Answer every question — there is no penalty for guessing.
5. Mark questions you are unsure of and return to them if time permits.
6. Do not consult reference materials during the timed simulation.
7. After completing, use the Answer Key (Section 3) to score your responses and study the rationales for every question you missed.

8. Transfer your scores to the Domain Analysis Worksheet (Section 4) to identify areas of strength and areas needing additional study.

Scoring

Each question is worth 1 point. The total possible score is **100 points** for the practice exam and **4 points** per unit quick-check. Use the guide below to interpret your practice-exam result:

Score Range	Percentage	Interpretation
90-100	90-100%	Excellent — well prepared; focus on any missed domains
75-89	75-89%	Proficient — review highlighted weak domains before sitting for the official exam
60-74	60-74%	Developing — targeted study of weaker domains recommended; retake in 2-4 weeks
Below 60	<60%	Needs Work — systematic chapter review recommended before retaking

Accessibility Note

If you require accommodations (extended time, read-aloud, large print, or alternative format), please refer to the Instructor Notes in Section 5 or consult your instructor before beginning.

Instructions for Instructors: How to Use This Assessment Bank

This bank is designed for flexible deployment across multiple instructional contexts. The 120 questions span all nine AVIXA AV Technologist Test domains and are calibrated to three levels of Bloom's Taxonomy (Remember, Apply,

Analyze/Evaluate), making them suitable for both formative and summative assessment.

Recommended Uses

- **Full Practice Exam:** Administer all 100 questions as a 90-minute summative practice exam at the end of the course. Score and return with rationales to guide final review before the official AVIXA exam.
- **Pre-Test / Post-Test:** Administer a 20–25 question subset (drawn by domain) at the beginning of the course as a pre-test, then re-administer the same subset at the end to measure growth. The Quick-Check Bank is designed specifically for this use.
- **Chapter Quizzes:** Select 8–12 questions from the relevant domain section to construct chapter quizzes. The AVIXA Domain and Chapter Reference fields on each question facilitate rapid sorting.
- **Exit Tickets:** Select 2–3 application- or analysis-level questions per lesson to use as exit tickets or warm-up checks.
- **Review Stations:** Print questions on cards sorted by domain; use in small-group review rotations.
- **LMS Import:** A full QTI 2.1-compatible export of this question bank is available in the online resource portal. Refer to your LMS documentation for import procedures (Canvas, Schoology, Moodle, Blackboard all support QTI 2.1).

Differentiation Guidance

- For **recall-level learners**, prioritize questions tagged *Bloom's: Remember* and *Understand*.
- For **advanced learners**, prioritize questions tagged *Bloom's: Apply, Analyze,* and *Evaluate*.
- Questions marked with **classroom-teacher scenarios** are especially appropriate for non-technical educators building AV literacy.

How This Exam Maps to the AVIXA AV Technologist Test

The AVIXA AV Technologist (AVT) certification is an entry-level credential validating foundational knowledge across core AV disciplines. The official exam contains approximately 100 questions drawn from the domain blueprint maintained by AVIXA. This study aid mirrors that domain structure and approximate question weighting so that your practice-exam experience closely simulates the official test format.

Important

This is a **study aid only**. It is not affiliated with, licensed by, or a substitute for the official AVIXA AV Technologist examination. Passing this practice exam does not confer certification. Visit **avixa.org** to register for the official exam, review the current domain blueprint, and access official study materials.

AVIXA AV Technologist Test Domain Blueprint

The table below shows each domain, the approximate percentage of questions on the official exam, the number of questions in this practice exam, and the corresponding textbook chapter(s).

Domain	Official Exam Weight (approx.)	Practice Exam Questions	Quick-Check Questions	Textbook Chapter(s)
AV Industry Overview & Fundamentals	10%	10 (Q1-Q10)	Unit 1, Q101-Q102	Ch. 1
Audio Systems	15%	15 (Q11-Q25)	Unit 1, Q103-Q104	Ch. 2
Video and Display Systems	15%	15 (Q26-Q40)	Unit 2, Q105-Q106	Ch. 3
Signal Flow and	15%	15 (Q41-Q55)	Unit 2, Q107-	Ch. 4

Domain	Official Exam Weight (approx.)	Practice Exam Questions	Quick-Check Questions	Textbook Chapter(s)
Connectivity			Q108	
AV Networking	10%	10 (Q56-Q65)	Unit 3, Q109-Q110	Ch. 5
Control Systems	10%	10 (Q66-Q75)	Unit 3, Q111-Q112	Ch. 6
Collaboration and UC Systems	10%	10 (Q76-Q85)	Unit 4, Q113-Q114	Ch. 7
AV Design and Integration	10%	10 (Q86-Q95)	Unit 4, Q115-Q116	Ch. 8-9
Maintenance and Troubleshooting	5%	5 (Q96-Q100)	Unit 5, Q117-Q120	Ch. 10
TOTALS	100%	100	20	Ch. 1-10

ISTE Standards Alignment

Questions in this bank are further tagged to relevant **ISTE Standards for Educators** where applicable. The primary alignments are:

- **ISTE Standard 2 - Leader:** Using technology to create equitable learning environments (AV design, accessibility, hybrid equity)
 - **ISTE Standard 3 - Citizen:** Digital citizenship in AV contexts, data privacy in UC systems
 - **ISTE Standard 5 - Designer:** Designing AV-rich learning environments using UDL principles
 - **ISTE Standard 6 - Facilitator:** Using AV tools to empower learner voice and foster engagement
-

Section 1: 100-Question Practice Exam



Exam Conditions — 90 Minutes Recommended

Select the single best answer for each question. Do not use reference materials during timed conditions. Answer all questions; there is no penalty for guessing. Mark uncertain questions and return to them as time allows.

Domain 1: AV Industry Overview & Fundamentals (Q1-Q10)

Q1. Which organization is primarily responsible for developing and publishing technical standards for the audiovisual industry?

- A. IEEE
- B. AVIXA
- C. ANSI
- D. TIA

AVIXA Domain: AV Industry Overview & Fundamentals | Chapter Reference: Ch. 1

Q2. A new educator at a middle school is asked to join the school's AV committee. She asks what "AV integration" means in an educational setting. Which of the following BEST describes AV integration?

- A. Replacing teachers with automated display systems
- B. Combining audio, video, and control technologies into a unified, functional system within a space
- C. Installing individual consumer electronics devices in each classroom
- D. Streaming educational videos from the internet to student devices

AVIXA Domain: AV Industry Overview & Fundamentals | Chapter Reference: Ch. 1

Q3. Which of the following BEST defines "signal" in the context of AV systems?

- A. Electrical energy used to power AV equipment
- B. A representation of information (audio, video, or data) carried by an electrical or optical medium
- C. A wireless transmission protocol used exclusively for video
- D. The cable connector type used to link AV devices

AVIXA Domain: AV Industry Overview & Fundamentals | Chapter Reference: Ch. 1

Q4. The AV industry uses the term "ecosystem" to describe the network of products, professionals, and standards that support AV solutions. Which role is MOST directly responsible for ensuring installed AV systems operate as designed after installation?

- A. AV designer
- B. AV technologist / systems technician
- C. AV salesperson
- D. Electrical contractor

AVIXA Domain: AV Industry Overview & Fundamentals | Chapter Reference: Ch. 1

Q5. An educator preparing a lesson on the history of AV technology asks how the transition from analog to digital signals changed the AV industry. Which of the following is the MOST significant advantage digital signals offer over analog signals?

- A. Digital signals require less bandwidth than analog signals
- B. Digital signals are immune to all forms of interference
- C. Digital signals can be copied, processed, and transmitted with minimal degradation
- D. Digital signals eliminate the need for amplification

AVIXA Domain: AV Industry Overview & Fundamentals | Chapter Reference: Ch. 1

Q6. Universal Design for Learning (UDL) principles encourage educators to provide multiple means of representation. Which AV system feature MOST directly supports UDL in a classroom?

- A. High-gain projection screens
- B. Assistive listening systems that transmit audio wirelessly to hearing-impaired students
- C. High-definition cameras for teacher surveillance
- D. Motorized screen lifts

AVIXA Domain: AV Industry Overview & Fundamentals | Chapter Reference: Ch. 1

Q7. Which of the following BEST describes the role of ISTE Standards in an AV-equipped learning environment?

- A. They specify the exact AV hardware required in each classroom type
- B. They provide a framework for how educators and students should use technology to enhance learning
- C. They replace local school district technology policies
- D. They are regulatory requirements enforced by federal law

AVIXA Domain: AV Industry Overview & Fundamentals | Chapter Reference: Ch. 1

Q8. A school district is planning a capital improvement project to upgrade AV systems in 12 classrooms. Which document is typically developed FIRST to guide the project scope, budget, and technical requirements?

- A. A purchase order
- B. A scope of work / needs assessment
- C. A cable schedule
- D. An as-built drawing

AVIXA Domain: AV Industry Overview & Fundamentals | Chapter Reference: Ch. 1

Q9. The term "AV over IP" (AVoIP) refers to which of the following?

- A. Replacing all AV equipment with internet-connected consumer devices
- B. Distributing audio and video signals across a standard network infrastructure using IP protocols

- C. Streaming publicly available content from the internet to classroom displays
- D. Controlling AV equipment using a smartphone app

AVIXA Domain: AV Industry Overview & Fundamentals | Chapter Reference: Ch. 1

Q10. Which of the following emerging technologies is MOST likely to change how AV content is distributed across large school campuses in the near future?

- A. VHS tape duplication
- B. Software-defined AV over IP with centralized management
- C. Coaxial cable distribution with analog switchers
- D. Stand-alone Blu-ray players in each room

AVIXA Domain: AV Industry Overview & Fundamentals | Chapter Reference: Ch. 1

Domain 2: Audio Systems (Q11-Q25)

Q11. A teacher notices that students in the back rows of her classroom frequently ask her to repeat herself. The most likely AV-related solution would involve which of the following?

- A. Installing additional projectors at the back of the room
- B. Implementing a classroom audio distribution system with ceiling-mounted speakers
- C. Replacing the display with a larger model
- D. Increasing the brightness of the display

AVIXA Domain: Audio Systems | Chapter Reference: Ch. 2

Q12. A decibel (dB) is best defined as which of the following?

- A. An absolute unit of sound pressure equal to 1 pascal
- B. A logarithmic ratio comparing a measured level to a reference level
- C. The frequency range of human hearing measured in hertz
- D. The impedance value of a speaker measured in ohms

AVIXA Domain: Audio Systems | Chapter Reference: Ch. 2

Q13. If an amplifier produces a signal that is 10 dB louder than its input, and the input is at 0 dBu, what is the output level?

- A. 0 dBu
- B. 5 dBu
- C. 10 dBu
- D. 100 dBu

AVIXA Domain: Audio Systems | Chapter Reference: Ch. 2

Q14. Which microphone type operates by generating a voltage when sound waves cause a conductive diaphragm to vibrate within a magnetic field?

- A. Condenser microphone
- B. Dynamic microphone
- C. Ribbon microphone
- D. Boundary microphone

AVIXA Domain: Audio Systems | Chapter Reference: Ch. 2

Q15. A condenser microphone requires phantom power to operate. What voltage does standard phantom power supply?

- A. 9V DC
- B. 24V DC
- C. 48V DC
- D. 120V AC

AVIXA Domain: Audio Systems | Chapter Reference: Ch. 2

Q16. A cardioid microphone polar pattern picks up sound PRIMARILY from which direction?

- A. All directions equally (omnidirectional)
- B. Two opposite directions (figure-eight)

- C. The front of the microphone, with rejection at the rear
- D. A very narrow beam directly in front (supercardioid only)

AVIXA Domain: Audio Systems | Chapter Reference: Ch. 2

Q17. During a school assembly, the audio technician notices a loud, sustained tone coming from the speakers when the presenter moves near them. What is the MOST likely cause?

- A. Low battery in the wireless microphone
- B. Acoustic feedback caused by the microphone picking up its own amplified output
- C. A broken XLR cable connector
- D. An incorrect impedance match between the mixer and amplifier

AVIXA Domain: Audio Systems | Chapter Reference: Ch. 2

Q18. Impedance in an audio system is measured in which unit?

- A. Decibels (dB)
- B. Hertz (Hz)
- C. Ohms (Ω)
- D. Watts (W)

AVIXA Domain: Audio Systems | Chapter Reference: Ch. 2

Q19. Human hearing spans approximately which frequency range?

- A. 20 Hz to 20,000 Hz
- B. 200 Hz to 2,000 Hz
- C. 100 Hz to 10,000 Hz
- D. 50 Hz to 50,000 Hz

AVIXA Domain: Audio Systems | Chapter Reference: Ch. 2

Q20. An ADA-compliant assistive listening system installed in a school auditorium ensures that users with hearing loss can receive audio signals directly to compatible devices. Which of the following systems is MOST commonly used for this purpose in educational spaces?

- A. FM assistive listening system
- B. AM radio broadcast system
- C. Bluetooth audio system
- D. Infrared (IR) illuminator system

AVIXA Domain: Audio Systems | Chapter Reference: Ch. 2

Q21. A school music director is choosing speakers for a recital hall. She wants the speakers to reproduce sound evenly across the entire audience area. Which speaker coverage specification should she prioritize?

- A. Maximum SPL rating
- B. Dispersion angle (coverage angle)
- C. Input impedance
- D. Frequency response of the subwoofer

AVIXA Domain: Audio Systems | Chapter Reference: Ch. 2

Q22. A balanced audio connection uses three conductors. What is the PRIMARY advantage of balanced audio over unbalanced audio in a school AV installation?

- A. It allows the signal to travel at twice the speed
- B. It provides common-mode noise rejection, reducing hum and interference over long cable runs
- C. It amplifies the signal without needing a separate amplifier
- D. It eliminates the need for phantom power

AVIXA Domain: Audio Systems | Chapter Reference: Ch. 2

Q23. A digital signal processor (DSP) in an AV system is primarily used to perform which function?

- A. Convert AC power to DC power for AV equipment
- B. Manipulate audio signals using algorithms for equalization, dynamics, delay, and routing
- C. Transmit video signals over fiber-optic cables
- D. Provide wireless connectivity between AV devices

Q24. An educator facilitating a hybrid class notices that remote participants report the teacher's voice sounds "boomy" or has too much bass. Which DSP tool would MOST effectively address this issue?

- A. A gate / expander
- B. A low-cut (high-pass) filter on the microphone channel
- C. A stereo bus compressor
- D. A delay line on the speaker feed

Q25. When placing ceiling speakers in a classroom to achieve even coverage, the AV designer typically uses which calculation to determine optimal speaker spacing?

- A. Speaker spacing = ceiling height $\times$ 2
- B. Speaker spacing based on the speaker's coverage angle and mounting height, often using the tangent function
- C. Speaker spacing = room length $\div$ number of students
- D. Speaker spacing = amplifier wattage $\div$ speaker impedance

Domain 3: Video and Display Systems (Q26-Q40)

Q26. A teacher notices that the projector image in her classroom appears washed out and faded during morning classes when sunlight enters through east-facing windows. Which projector specification is MOST relevant to addressing this problem?

- A. Contrast ratio
- B. Lumens (brightness) output
- C. Native resolution
- D. Aspect ratio

Q27. The throw ratio of a projector is calculated using which formula?

- A. Throw ratio = Screen width ÷ Throw distance
- B. Throw ratio = Throw distance ÷ Screen width
- C. Throw ratio = Lumens ÷ Screen area
- D. Throw ratio = Resolution ÷ Pixel density

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Q28. A school library has limited space between the projector mounting location and the screen. Which type of projector would BEST address this physical constraint?

- A. Standard-throw projector
- B. Long-throw projector
- C. Short-throw or ultra-short-throw projector
- D. Rear-projection system

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Q29. What does the term "native resolution" mean when describing a display device?

- A. The maximum resolution the display can accept from an external source
- B. The fixed number of physical pixels in the display panel
- C. The resolution recommended by the manufacturer for optimal color accuracy
- D. The resolution at which the display operates when connected to older devices

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Q30. A 4K UHD display has a native resolution of how many horizontal × vertical pixels?

- A. 1920 × 1080
- B. 2560 × 1440
- C. 3840 × 2160
- D. 7680 × 4320

Q31. An educator is designing a collaborative learning space and must choose between a large-format display (LCD flat panel) and a projector for a room with ambient light. Which factor MOST favors the flat-panel LCD display?

- A. Lower total cost of ownership over 5 years
- B. Ability to display a larger image in all conditions
- C. Superior brightness and contrast in ambient light conditions
- D. Portability between rooms

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Q32. What does "contrast ratio" describe in a display device?

- A. The ratio of the display's maximum brightness to its minimum brightness (black level)
- B. The ratio of horizontal pixels to vertical pixels
- C. The color temperature range the display can reproduce
- D. The difference in refresh rate between two display standards

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Q33. A digital signage system in a school hallway must display content that meets WCAG 2.1 Level AA accessibility guidelines. Which of the following design choices BEST supports this requirement?

- A. Using text in light gray on a white background
- B. Ensuring a minimum contrast ratio of 4.5:1 between text and background for normal-sized text
- C. Auto-cycling content every 2 seconds
- D. Using animated GIFs without captions

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Q34. The aspect ratio 16:9 is MOST commonly associated with which display application?

- A. Traditional 35mm film projection
- B. HD and UHD video content and modern widescreen displays
- C. Standard-definition 4:3 television
- D. Digital cinema (DCI) 2.39:1 scope format

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Q35. LED display technology used in direct-view video walls differs from traditional LCD flat panels primarily because:

- A. LEDs cannot reproduce color accurately at large sizes
- B. Direct-view LED uses individual LED modules as pixels rather than a backlit LCD panel, enabling seamless tiling and very high brightness
- C. Direct-view LED requires analog input signals only
- D. LED video walls consume less power than a single LCD display of equivalent size

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Q36. A teacher in a science classroom reports that text on the projected screen appears fuzzy and difficult to read, even though the computer source is set to 1920×1080. The projector's native resolution is 1024×768. What is the MOST likely cause of the poor image quality?

- A. The projector lamp is near end of life
- B. The source resolution does not match the projector's native resolution, causing image scaling artifacts
- C. The HDMI cable is too long
- D. The projector's contrast ratio is too low

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Q37. An interactive flat-panel display (IFPD) combines which two technologies?

- A. A projector and a whiteboard
- B. A large-format display and a touch-sensitive input layer

- C. A document camera and a streaming receiver
- D. A LED display and an integrated audio amplifier only

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Q38. When planning display placement for a classroom, the AVIXA recommended maximum viewing distance for a standard-definition display is generally calculated as how many times the screen height?

- A. 2× screen height
- B. 4× screen height
- C. 6× screen height
- D. 8× screen height

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Q39. A high school librarian wants to set up a digital signage display that automatically updates with school news. Which component is essential for delivering dynamic, remotely updated content to the display?

- A. A stand-alone Blu-ray player
- B. A digital signage media player connected to a content management system (CMS)
- C. A document camera with HDMI output
- D. An analog cable television tuner

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Q40. What is "keystone correction" in a projector, and when is it typically needed?

- A. A color calibration feature used when projecting in dark rooms
- B. A digital or optical adjustment that corrects the trapezoidal image distortion caused when the projector is not perpendicular to the screen
- C. A lens feature that increases image brightness at the edges
- D. A setting that adjusts the projector's native resolution to match the source signal

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Domain 4: Signal Flow and Connectivity (Q41-Q55)

Q41. Signal flow in an AV system describes which of the following?

- A. The route electrical power takes from the outlet to each device
- B. The path audio and/or video signals travel from a source, through processing and distribution, to an output destination
- C. The physical route of cables through conduit in a building
- D. The sequence in which AV devices power on at system startup

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q42. Which of the following cable types carries both digital audio and video signals on a single connector and is the MOST common connection between a laptop and a classroom display?

- A. VGA (DE-15)
- B. XLR
- C. HDMI
- D. RS-232

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q43. What is the PRIMARY difference between HDMI and DisplayPort in terms of AV signal delivery?

- A. HDMI is an analog standard; DisplayPort is digital
- B. DisplayPort uses a packet-based protocol better suited for high-refresh-rate and multi-stream (daisy-chain) configurations, while HDMI is more prevalent in consumer electronics
- C. DisplayPort cannot carry audio signals
- D. HDMI supports higher resolutions than DisplayPort in all versions

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q44. A teacher's laptop only has a USB-C port. Which adapter or cable type would allow her to connect directly to an HDMI display in her classroom?

- A. USB-C to VGA adapter
- B. USB-C to HDMI adapter (using USB-C with DisplayPort Alt Mode or Thunderbolt)
- C. USB-C to RS-232 serial adapter
- D. USB-C to XLR adapter

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q45. An AV installer is running signal cables for a new lecture hall. She needs to carry a high-definition video signal over a distance of 100 meters without signal degradation. Which cable type is MOST appropriate?

- A. Standard HDMI copper cable
- B. Composite video (RCA) coaxial cable
- C. Fiber-optic HDMI or HDBaseT over fiber
- D. Standard Cat 5e UTP cable without any extender

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q46. HDBaseT technology is designed to transmit which combination of signals over a single Cat cable?

- A. Audio only, up to 4K resolution equivalent
- B. Uncompressed HD/4K video, audio, control (RS-232 and IR), USB, and PoE (power over Ethernet) over a single Cat 5e/6 cable
- C. Compressed video using H.264 codec over a Cat 6 network cable
- D. Video and power only, with a maximum run of 30 meters

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q47. A technician notices that a 4K signal degrades to 1080p after passing through a long HDMI cable run. What is the MOST likely cause?

- A. The display does not support 4K resolution
- B. The HDMI cable is not rated for the bandwidth required by 4K (18 Gbps), or the cable run exceeds the standard's passive cable distance limit

- C. The source device is outputting the wrong aspect ratio
- D. The HDMI ARC channel is enabled

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q48. In the context of AV systems, what is the purpose of an AV matrix switcher?

- A. To convert analog signals to digital format
- B. To route any input signal to any output destination independently and simultaneously
- C. To amplify weak video signals for long cable runs
- D. To mix multiple audio channels into a stereo output

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q49. HDCP (High-bandwidth Digital Content Protection) is an encryption standard applied to HDMI and DisplayPort signals. A teacher reports that her laptop display appears on her monitor but not on the classroom projector. HDCP compliance issues could cause this if:

- A. The projector is not HDCP-compliant and the source is transmitting protected content
- B. The HDMI cable is too short
- C. The projector's lamp is at low brightness
- D. The computer's display adapter has too many pixels

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q50. Which connector type is used for professional balanced audio connections and features three pins (ground, hot, cold)?

- A. RCA (phono)
- B. 3.5 mm TRS
- C. XLR
- D. TS (mono phone plug)

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q51. A school AV technician is labeling cables in a newly installed system. Which practice is considered the industry standard for cable labeling?

- A. Labeling only at the point of origin
- B. Labeling at both ends of every cable with consistent identifiers that match the as-built documentation
- C. Color-coding only, without alphanumeric labels
- D. Labeling cables only when they cross building infrastructure (walls, ceilings)

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q52. What is the function of a DA (distribution amplifier) in an AV system?

- A. To convert digital audio signals to analog
- B. To split and amplify a single AV signal to feed multiple outputs simultaneously without signal loss
- C. To switch between multiple input sources
- D. To provide phantom power to condenser microphones

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q53. An educator notices a "ground loop hum" in the audio system when she connects her laptop to the AV system. The MOST effective solution is typically to:

- A. Replace the HDMI cable with a longer one
- B. Insert a ground loop isolator or a DI (direct injection) box with a ground-lift switch
- C. Increase the gain on the audio mixer
- D. Connect the laptop to a different USB port

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q54. The EDID (Extended Display Identification Data) protocol is used in HDMI systems to accomplish which task?

- A. Encrypt the video signal to prevent unauthorized viewing
- B. Allow the source device to query the display's supported resolutions, refresh rates, and audio formats, enabling automatic format negotiation
- C. Provide power to the display over the HDMI cable
- D. Control display brightness and color temperature remotely

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q55. A teacher's presentation laptop uses a wireless presentation system to share content to the classroom display. Which of the following is a potential AV connectivity concern specific to wireless presentation systems?

- A. Wireless systems always produce higher image quality than wired HDMI
- B. Latency, compression artifacts, and potential network congestion on shared Wi-Fi infrastructure may degrade the experience
- C. Wireless systems cannot carry audio to the AV system
- D. Wireless systems are incompatible with 4K resolution at any compression level

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Domain 5: AV Networking (Q56-Q65)

Q56. In AV over IP systems, network bandwidth is a critical concern. Uncompressed 1080p60 video requires approximately how much bandwidth?

- A. 10 Mbps
- B. 100 Mbps
- C. 1 Gbps
- D. 3 Gbps or more

AVIXA Domain: AV Networking | Chapter Reference: Ch. 5

Q57. Which network protocol is used to assign IP addresses automatically to devices on a network, including AV endpoints?

- A. DNS (Domain Name System)
- B. DHCP (Dynamic Host Configuration Protocol)
- C. SNMP (Simple Network Management Protocol)
- D. NTP (Network Time Protocol)

AVIXA Domain: AV Networking | Chapter Reference: Ch. 5

Q58. A school district IT director wants to ensure that AV-over-IP traffic does not congest the general-purpose data network used by students for internet access. Which network strategy BEST achieves this goal?

- A. Installing a separate physical cable for every AV device
- B. Segmenting AV traffic onto a dedicated VLAN and applying QoS policies
- C. Reducing the resolution of all AV streams to 480p
- D. Using only analog signal distribution throughout the campus

AVIXA Domain: AV Networking | Chapter Reference: Ch. 5

Q59. Quality of Service (QoS) in a network context refers to which of the following?

- A. The physical quality of network cables measured in TIA categories
- B. The ability to prioritize specific types of traffic (such as AV streams) to ensure consistent performance and low latency
- C. The security encryption standard used on wireless networks
- D. The maximum number of devices a switch can support

AVIXA Domain: AV Networking | Chapter Reference: Ch. 5

Q60. In an IPv4 addressing scheme, which address range is reserved for private network use (not routable on the public internet)?

- A. 200.0.0.0 – 200.255.255.255
- B. 192.168.0.0 – 192.168.255.255 (one of three private ranges)
- C. 100.0.0.0 – 100.255.255.255
- D. 8.8.8.0 – 8.8.8.255

AVIXA Domain: AV Networking | Chapter Reference: Ch. 5

Q61. Multicast transmission in AV-over-IP networks is preferred over unicast for multi-room distribution because:

- A. Multicast requires a dedicated fiber connection for each receiver
- B. Multicast sends a single stream that multiple receivers can join, reducing total bandwidth consumption compared to sending separate unicast streams to each receiver
- C. Multicast eliminates the need for a managed network switch
- D. Multicast converts uncompressed video to JPEG 2000 automatically

AVIXA Domain: AV Networking | Chapter Reference: Ch. 5

Q62. A managed network switch is required for AV-over-IP systems primarily because:

- A. Unmanaged switches cannot handle Ethernet frames at any speed
- B. Managed switches support VLAN configuration, QoS tagging, IGMP snooping (for multicast control), and traffic monitoring essential for AV performance
- C. Managed switches amplify the AV signal for long cable runs
- D. Managed switches convert IP signals to HDMI format

AVIXA Domain: AV Networking | Chapter Reference: Ch. 5

Q63. Which of the following network security practices is MOST important when deploying AV devices on a school network?

- A. Enabling the default administrator credentials on all AV devices for easy access
- B. Changing default credentials, applying firmware updates, and placing AV devices on isolated network segments
- C. Disabling all firewall rules to allow AV traffic to flow freely
- D. Using only consumer-grade Wi-Fi access points for AV distribution

AVIXA Domain: AV Networking | Chapter Reference: Ch. 5

Q64. IGMP snooping is a managed switch feature relevant to AV-over-IP systems because it:

- A. Encrypts multicast video streams to prevent unauthorized viewing
- B. Allows the switch to forward multicast streams only to ports where receivers have subscribed, preventing multicast flooding to all ports
- C. Converts multicast streams to unicast for legacy devices
- D. Monitors network jitter and corrects packet timing errors

AVIXA Domain: AV Networking | Chapter Reference: Ch. 5

Q65. A school AV technician is troubleshooting an AV-over-IP system in which video streams drop intermittently. After confirming the network switch is managed and QoS is configured, what should be checked NEXT?

- A. Whether all AV endpoints have static or properly reserved IP addresses, and that no IP address conflicts exist on the network
- B. Whether the display's native resolution matches the projector lamp brightness
- C. Whether the audio channels are balanced or unbalanced
- D. Whether the room's acoustic treatment is adequate

AVIXA Domain: AV Networking | Chapter Reference: Ch. 5

Domain 6: Control Systems (Q66-Q75)

Q66. An AV control system in a classroom serves which PRIMARY function?

- A. Amplifying audio signals from microphones
- B. Providing a centralized, programmable interface that allows users to operate multiple AV devices through a single interface or touch panel
- C. Converting analog video signals to digital for network distribution
- D. Supplying power to AV devices through PoE

AVIXA Domain: Control Systems | Chapter Reference: Ch. 6

Q67. Which of the following control protocols is commonly used by a control system processor to send serial commands to projectors and displays?

- A. HDMI CEC
- B. RS-232 serial communication

C. DMX-512

D. MIDI

AVIXA Domain: Control Systems | Chapter Reference: Ch. 6

Q68. A teacher at an elementary school reports that the AV system's touch panel is confusing for substitute teachers and students. From a UI design perspective, which improvement would MOST effectively reduce user errors?

- A. Adding more buttons to the touch panel to expose all system functions
- B. Using small text labels to conserve screen space
- C. Simplifying the interface to display only the most commonly used functions, using clear icons and labels appropriate for the user's skill level
- D. Replacing the touch panel with a command-line interface

AVIXA Domain: Control Systems | Chapter Reference: Ch. 6

Q69. Infrared (IR) control in AV systems works by:

- A. Sending encoded light pulses in the IR spectrum from a blaster to a device's IR receiver
- B. Using radio-frequency signals to communicate over walls and obstructions
- C. Transmitting control data over the Ethernet network
- D. Using RS-485 serial protocol for bidirectional communication

AVIXA Domain: Control Systems | Chapter Reference: Ch. 6

Q70. A control system sends an RS-232 command to power on a display, but the display does not respond. The technician has verified the RS-232 cable is properly connected. What should be checked NEXT?

- A. The display's HDMI input selection
- B. The baud rate, data bits, stop bits, and parity settings — which must match between the control processor and the display
- C. The projector lamp life counter
- D. The DHCP lease on the network switch

AVIXA Domain: Control Systems | Chapter Reference: Ch. 6

Q71. In a room scheduling and occupancy context, which control system feature allows AV equipment to power down automatically when a room is unoccupied, reducing energy consumption?

- A. Auto-scaling display output
- B. Occupancy sensor integration with control system logic (e.g., auto-shutoff after inactivity)
- C. EDID management
- D. QoS traffic prioritization

AVIXA Domain: Control Systems | Chapter Reference: Ch. 6

Q72. API (Application Programming Interface) integration allows a control system to:

- A. Connect physical XLR cables without adapters
- B. Communicate with external software platforms (such as room booking systems, calendar apps, or UC platforms) through standardized data exchange
- C. Convert analog audio signals to AES/EBU digital format
- D. Increase the bandwidth of HDMI signals

AVIXA Domain: Control Systems | Chapter Reference: Ch. 6

Q73. A school district wants teachers to be able to start a classroom AV session (display on, source selected, volume set) with a single button press. This is BEST accomplished through which control system feature?

- A. A preset macro that executes a programmed sequence of commands automatically
- B. Manual adjustment of each device individually via IR remote controls
- C. A hardware relay that toggles the power strip
- D. The display's built-in OSD (on-screen display) menu

AVIXA Domain: Control Systems | Chapter Reference: Ch. 6

Q74. Which of the following is a potential cybersecurity concern when a control system processor is connected to the school's IP network?

- A. The control processor may reduce the display's lumens output
- B. Unauthorized access to the control system could allow an intruder to manipulate AV devices, disable displays, or use the system as a network entry point
- C. Network connectivity causes audio feedback in the speaker system
- D. IP-connected control systems always require a dedicated fiber connection

AVIXA Domain: Control Systems | Chapter Reference: Ch. 6

Q75. A control system programmer is writing logic to automate window shading and lighting adjustments when the projector is activated in a classroom. This type of integration is often referred to as:

- A. Signal flow mapping
- B. Environmental integration or "smart room" automation
- C. EDID negotiation
- D. Codec transcoding

AVIXA Domain: Control Systems | Chapter Reference: Ch. 6

Domain 7: Collaboration and UC Systems (Q76-Q85)

Q76. A teacher is setting up a hybrid class in which some students are in the classroom and others are joining remotely via video conferencing. She notices that remote participants cannot hear the in-room students. What is the MOST likely cause?

- A. The display resolution is too low
- B. The room microphone is not routed to the video conferencing system's audio input
- C. The HDMI cable from the laptop to the display is faulty
- D. The remote participants' internet connection is too slow

AVIXA Domain: Collaboration and UC Systems | Chapter Reference: Ch. 7

Q77. "Hybrid equity" in a blended learning environment refers to which principle?

- A. Providing equal numbers of in-room and remote seats in every class session
- B. Designing AV and UC systems so that remote participants have an equivalent learning experience to in-room participants — including audio clarity, visual access, and interactive participation
- C. Ensuring all students use the same video conferencing platform
- D. Requiring remote participants to have high-speed broadband internet

AVIXA Domain: Collaboration and UC Systems | Chapter Reference: Ch. 7

Q78. A "codec" in a video conferencing system refers to which component?

- A. The physical cable connecting the camera to the display
- B. A hardware or software system that compresses (encodes) and decompresses (decodes) audio and video signals for transmission
- C. The remote control unit for a conference room display
- D. The power supply for a UC endpoint

AVIXA Domain: Collaboration and UC Systems | Chapter Reference: Ch. 7

Q79. Which of the following video conferencing camera features is MOST beneficial for a classroom where the teacher moves around the room while presenting?

- A. A fixed wide-angle lens with no pan/tilt capability
- B. Auto-framing / presenter tracking camera with pan, tilt, and zoom capability
- C. A document camera positioned above the teacher's desk
- D. A 4K webcam mounted on the teacher's laptop

AVIXA Domain: Collaboration and UC Systems | Chapter Reference: Ch. 7

Q80. Acoustic echo cancellation (AEC) in a UC system is designed to prevent which problem?

- A. Background noise from HVAC systems entering the microphone
- B. The far-end participants' audio being picked up by the room microphone and fed back to them as an echo

- C. Keystone distortion on the display
- D. IP address conflicts between UC endpoints

AVIXA Domain: Collaboration and UC Systems | Chapter Reference: Ch. 7

Q81. A school district is selecting a video conferencing platform for all classrooms. The IT coordinator needs to ensure it is compatible with the existing room AV systems. Which technical factor is MOST important to verify?

- A. The platform's marketing cost per seat
- B. Whether the platform supports standard protocols (SIP/H.323) or has certified hardware integrations with the installed room AV equipment
- C. The platform's social media sharing features
- D. Whether the platform supports analog audio connections only

AVIXA Domain: Collaboration and UC Systems | Chapter Reference: Ch. 7

Q82. A teacher facilitating a hybrid discussion notices that when multiple in-room students speak at once, the system only transmits one voice to remote participants. This is a characteristic of which audio processing feature?

- A. Automatic gain control (AGC)
- B. Voice activity detection (VAD) with single-talker priority — often an unintended limitation in some AEC/mixing configurations
- C. Acoustic isolation
- D. Noise gate set too low

AVIXA Domain: Collaboration and UC Systems | Chapter Reference: Ch. 7

Q83. Which of the following BEST describes the purpose of a "content camera" or document camera in a hybrid classroom?

- A. To provide wide-angle coverage of the entire classroom for security purposes
- B. To allow teachers to share physical documents, objects, or hand-written notes as live video with both in-room and remote participants

- C. To project slides from the teacher's laptop to the classroom display
- D. To record the class session for asynchronous playback only

AVIXA Domain: Collaboration and UC Systems | Chapter Reference: Ch. 7

Q84. From an ISTE Standard 2 (Leader) perspective, which hybrid classroom AV design choice BEST promotes an equitable learning environment for students with disabilities?

- A. Installing a 4K camera with no microphone system
- B. Including closed captioning display, assistive listening connectivity, and accessible UC platform features alongside the standard AV installation
- C. Providing remote participants with a dedicated display but no audio connection
- D. Using only proprietary protocols to secure the video conference

AVIXA Domain: Collaboration and UC Systems | Chapter Reference: Ch. 7

Q85. A bandwidth-limited school with a shared 100 Mbps internet connection needs to support 10 simultaneous video conference sessions. Approximately how much bandwidth should be budgeted per session to maintain acceptable HD video quality (720p30)?

- A. 100 Kbps per session
- B. 1-3 Mbps per session (up and down)
- C. 10 Mbps per session
- D. 50 Mbps per session

AVIXA Domain: Collaboration and UC Systems | Chapter Reference: Ch. 7

Domain 8: AV Design and Integration (Q86-Q95)

Q86. During the AV design process, a "needs assessment" is conducted primarily to:

- A. Determine the total wattage of electrical panels in the building
- B. Identify the functional requirements, workflows, user needs, and constraints that will inform the AV system design

- C. Select the specific equipment brands to be installed
- D. Develop the final as-built drawings

AVIXA Domain: AV Design and Integration | Chapter Reference: Ch. 8-9

Q87. When designing sightlines in a classroom or lecture hall, the AV designer must ensure that all audience members have an unobstructed view of the display. AVIXA guidelines recommend that the MINIMUM viewing distance from a screen be no closer than what multiple of the screen height?

- A. 0.5× screen height
- B. 1× screen height
- C. 1.5× screen height
- D. 3× screen height

AVIXA Domain: AV Design and Integration | Chapter Reference: Ch. 8-9

Q88. The bottom edge of a classroom projection screen should be mounted at a minimum height above the floor. Which guideline is MOST commonly referenced for this measurement?

- A. The screen bottom should be at least 1.2 meters (48 inches) above the finished floor to maintain sightlines over seated audience members' heads
- B. The screen bottom should be at the teacher's eye level
- C. The screen bottom should be flush with the whiteboard top
- D. The screen bottom should be at the front-row student's eye level when standing

AVIXA Domain: AV Design and Integration | Chapter Reference: Ch. 8-9

Q89. Room acoustics play a critical role in AV system performance. A classroom with many hard, parallel surfaces (concrete walls, tile floors) will suffer from which acoustic problem?

- A. Too little bass (bass deficiency)
- B. Long reverberation time (excessive echo), which reduces speech intelligibility
- C. Complete absence of ambient noise
- D. Increased amplifier gain requirements only

Q90. An AV integrator is planning cable pathways in a school renovation. Which document describes where AV cables will run through walls, ceilings, and conduit to ensure coordination with electrical and structural contractors?

- A. The equipment rack elevation drawing
- B. The signal flow diagram
- C. The AV floor plan / infrastructure drawing showing cable pathways and device locations
- D. The equipment specification sheet

Q91. ADA (Americans with Disabilities Act) standards require that AV controls such as wall-mounted touch panels or button panels in schools be installed within a specific height range. Which height range is specified for side reach over an obstruction?

- A. 15 to 30 inches AFF (above finished floor)
- B. 15 to 48 inches AFF
- C. 36 to 60 inches AFF
- D. 48 to 72 inches AFF

Q92. A teacher working in a science lab with specialized equipment asks the AV designer to incorporate a document camera above the demonstration bench. This type of AV addition is best described as part of which design phase?

- A. System commissioning
- B. Functional design / user workflow accommodation
- C. Preventive maintenance planning
- D. Network infrastructure provisioning

Q93. Rack unit (U) is a measurement used in AV equipment racks. One rack unit (1U) equals which height?

- A. 1 inch
- B. 1.75 inches (44.45 mm)
- C. 2 inches
- D. 3.5 inches

AVIXA Domain: AV Design and Integration | Chapter Reference: Ch. 8-9

Q94. System commissioning in an AV integration project refers to which process?

- A. The physical installation of cables and equipment
- B. The process of verifying, testing, adjusting, and documenting that all installed AV systems meet design specifications and perform as intended before handover to the client
- C. The initial needs assessment and site survey
- D. The procurement of equipment from the manufacturer

AVIXA Domain: AV Design and Integration | Chapter Reference: Ch. 8-9

Q95. A principal wants the new media center AV system to be expandable as technology evolves without requiring a complete re-installation. Which design principle BEST ensures future scalability?

- A. Installing the lowest-cost equipment available today
- B. Using proprietary, single-vendor systems throughout
- C. Designing with open standards, modular architecture, and adequate conduit/cable capacity for future upgrades
- D. Selecting maximum-resolution displays regardless of current content needs

AVIXA Domain: AV Design and Integration | Chapter Reference: Ch. 8-9

Domain 9: Maintenance and Troubleshooting (Q96-Q100)

Q96. A projector in a classroom has begun to display a message indicating that the lamp is approaching end of life. A teacher asks if she should wait until the lamp fails before replacing it. The BEST practice is to:

- A. Wait until the lamp fails completely to maximize its lifespan
- B. Replace the lamp proactively before it fails, as recommended by the manufacturer's hours-based maintenance schedule, to prevent unexpected instructional downtime
- C. Clean the lamp with a damp cloth to restore brightness
- D. Increase the projector's fan speed to extend lamp life indefinitely

AVIXA Domain: Maintenance and Troubleshooting | Chapter Reference: Ch. 10

Q97. A structured troubleshooting methodology in AV systems typically begins with which step?

- A. Replacing the most expensive component first
- B. Gathering information — identifying and defining the problem clearly by observing symptoms, reviewing recent changes, and consulting system documentation
- C. Rebooting all AV devices simultaneously
- D. Calling the equipment manufacturer immediately

AVIXA Domain: Maintenance and Troubleshooting | Chapter Reference: Ch. 10

Q98. A teacher reports that the classroom AV system produces audio but no video. The AV technician has verified the display is powered on and the correct input is selected. What is the MOST logical next troubleshooting step using a systematic approach?

- A. Replace the display immediately
- B. Test the video signal path by substituting a known-good cable and/or source, isolating the fault to a specific component or cable
- C. Upgrade the control system firmware
- D. Power cycle the building's electrical panel

Q99. Preventive maintenance for AV systems in schools typically includes which of the following on a scheduled basis?

- A. Replacing all cables annually regardless of condition
- B. Cleaning projector filters, checking lamp hours, inspecting cable connections, testing control system functionality, and updating firmware
- C. Rebuilding the control system program from scratch each school year
- D. Replacing all wireless microphone batteries only when they stop working

AVIXA Domain: Maintenance and Troubleshooting | Chapter Reference: Ch. 10

Q100. A school's AV technician documents all troubleshooting steps taken, the solution found, and any parts replaced after resolving a system issue. This documentation practice is valuable PRIMARILY because:

- A. It satisfies equipment warranty requirements that mandate written service logs
- B. It creates an institutional knowledge base that helps future technicians diagnose recurring issues faster and provides records for warranty and maintenance contracts
- C. It is required by AVIXA certification to maintain AVT status
- D. It eliminates the need for preventive maintenance scheduling

AVIXA Domain: Maintenance and Troubleshooting | Chapter Reference: Ch. 10

Section 2: 20-Question Quick-Check Bank

Use these short formative assessments after completing each unit. Allow approximately 8–10 minutes per unit check. Each unit check contains 4 questions aligned to 2–3 chapters from the course.

✓ Formative Assessment Purpose

Quick-checks are intended as low-stakes self-assessment tools. Review your answers against the Answer Key in Section 3 (Q101-Q120) and note any areas of uncertainty for focused review before the full practice exam.

Unit 1 Quick-Check — Chapters 1-2: AV Fundamentals & Audio Systems

UNIT 1

Q101. Which term describes the combination of hardware, software, and professional services that constitute the complete AV industry ecosystem?

- A. AV signal chain
- B. AV infrastructure
- C. AV industry value chain
- D. AV protocol stack

AVIXA Domain: AV Industry Overview & Fundamentals | Chapter Reference: Ch. 1

Q102. An educator is reviewing her school's technology plan and finds a reference to "ADA Section 508 compliance" for AV systems. What does this compliance requirement primarily address?

- A. Electrical safety standards for AV equipment
- B. Ensuring AV and information technology systems are accessible to people with disabilities
- C. Minimum network bandwidth requirements for AV over IP
- D. Fire-suppression requirements for AV equipment racks

AVIXA Domain: AV Industry Overview & Fundamentals | Chapter Reference: Ch. 1

Q103. A teacher is presenting in a gymnasium. She is using a wireless hand-held microphone but notices the audio drops in and out when she walks to certain areas of the floor. What is the MOST likely cause?

- A. The microphone capsule is a condenser type
- B. RF interference or multipath dropouts in the wireless microphone's radio frequency transmission
- C. Phantom power has been enabled on the receiver
- D. The speaker impedance is mismatched

AVIXA Domain: Audio Systems | Chapter Reference: Ch. 2

Q104. When two audio signals are said to be "out of phase," what problem does this cause when they are combined?

- A. The combined signal becomes twice as loud
- B. The signals cancel each other out (partially or fully), causing reduced bass and a thin, hollow sound
- C. The frequency response shifts toward the treble frequencies only
- D. The impedance of the combined signal doubles

AVIXA Domain: Audio Systems | Chapter Reference: Ch. 2

Unit 2 Quick-Check — Chapters 3-4: Video/Display Systems & Signal Flow

UNIT 2

Q105. A teacher is preparing a video presentation for a school event and wants to ensure the video will display correctly on the 16:9 auditorium screen without black bars or stretching. She should ensure her presentation is created at which resolution?

- A. 800 × 600 (4:3)
- B. 1920 × 1080 (16:9)
- C. 1024 × 768 (4:3)
- D. 2560 × 1080 (ultra-wide)

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Q106. A teacher notices pixel-level distortion and color bleeding on the classroom display after several years of use. These symptoms are MOST consistent with which display issue?

- A. HDMI cable incompatibility
- B. Image burn-in (phosphor or pixel retention), common in static-content applications on OLED or plasma displays
- C. Projector keystone misalignment
- D. Source device resolution mismatch

AVIXA Domain: Video and Display Systems | Chapter Reference: Ch. 3

Q107. In a school AV system, the terms "source," "switcher," "amplifier," and "display" describe which conceptual model?

- A. The OSI model of network communication
- B. The signal flow chain: signal originates at a source, is selected/switched, processed/amplified, and delivered to an output
- C. The ISTE framework for technology integration
- D. The AV rack wiring diagram

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Q108. A school has installed a wireless HDMI extender to eliminate cables across a walkway. Teachers complain of occasional video freezing. What is the MOST likely cause?

- A. The display's native resolution is too high
- B. RF interference or physical obstructions in the 5 GHz transmission path of the wireless HDMI system
- C. The HDCP version on the source is outdated
- D. The projector screen material reflects too much ambient light

AVIXA Domain: Signal Flow and Connectivity | Chapter Reference: Ch. 4

Unit 3 Quick-Check — Chapters 5-6: AV Networking & Control Systems

UNIT 3

Q109. A school's AV-over-IP system uses a 1 Gbps network switch. The district wants to add four 4K uncompressed video streams simultaneously. Why might this create a problem?

- A. 4K video cannot be transmitted over Ethernet
- B. Uncompressed 4K video requires approximately 12 Gbps per stream, far exceeding the 1 Gbps switch capacity for even a single stream
- C. The network switch does not support multicast
- D. IP cameras cannot output 4K resolution

AVIXA Domain: AV Networking | Chapter Reference: Ch. 5

Q110. A school's AV technician notices that a networked AV device has been assigned the IP address 169.254.x.x. What does this address indicate?

- A. The device has been assigned a valid static IP address
- B. The device failed to receive an IP address from a DHCP server and has assigned itself an APIPA (Automatic Private IP Addressing) address, indicating a DHCP or network connectivity problem
- C. The device is using IPv6 addressing
- D. The device is in a secure VLAN segment

AVIXA Domain: AV Networking | Chapter Reference: Ch. 5

Q111. A newly installed control system touch panel shows the correct interface but none of the buttons control the AV equipment. The control processor is powered on. What is the MOST likely cause to investigate first?

- A. The display's HDMI cable is disconnected
- B. The control processor's IP address or RS-232 communication settings are incorrect, or the control program has not been uploaded to the processor

- C. The room's ambient temperature is too high
- D. The touch panel requires a firmware downgrade

AVIXA Domain: Control Systems | Chapter Reference: Ch. 6

Q112. A control system uses a relay to control a motorized projector screen. When the teacher presses "Projector On," the screen should lower. Currently the projector turns on but the screen does not move. What is the MOST likely cause?

- A. The projector lamp needs replacement
- B. The relay trigger command is not being sent, the relay wiring is incorrect, or the relay contact is damaged — preventing the screen motor circuit from closing
- C. The HDMI source has the wrong aspect ratio selected
- D. The control processor needs a network cable upgrade

AVIXA Domain: Control Systems | Chapter Reference: Ch. 6

Unit 4 Quick-Check — Chapters 7-8: Collaboration/UC Systems & AV Design

UNIT 4

Q113. In a hybrid classroom AV design, the "camera placement problem" refers to which challenge?

- A. Finding a camera with sufficient zoom range for outdoor use
- B. Positioning the classroom camera so that remote participants can see the teacher and in-room students clearly, while the in-room students can make natural eye contact with the remote participants on the display
- C. Selecting a camera that matches the display's color profile
- D. Mounting the camera high enough so it is out of students' reach

AVIXA Domain: Collaboration and UC Systems | Chapter Reference: Ch. 7

Q114. Noise cancellation technology in UC microphone systems uses which process to reduce background noise?

- A. A physical noise barrier installed in front of the microphone
- B. DSP algorithms (and increasingly AI-based models) that distinguish speech from non-speech sounds and attenuate background noise in real time
- C. A directional speaker aimed at the microphone to produce anti-noise
- D. Reducing the microphone's gain until only close-range speech is captured

AVIXA Domain: Collaboration and UC Systems | Chapter Reference: Ch. 7

Q115. During an AV design site survey for a new classroom, the designer measures a reverberation time (RT60) that is too long for effective speech intelligibility. What RT60 value is generally recommended for a standard classroom?

- A. RT60 < 0.2 seconds
- B. RT60 between 0.4 and 0.6 seconds
- C. RT60 between 1.5 and 2.0 seconds
- D. RT60 > 3.0 seconds

AVIXA Domain: AV Design and Integration | Chapter Reference: Ch. 8-9

Q116. An AV designer is calculating the required screen size for a classroom that is 12 meters deep. Using the AVIXA guideline that the maximum viewing distance is approximately 6× the screen height for standard-definition content, what is the minimum screen height required?

- A. 1 meter
- B. 2 meters
- C. 3 meters
- D. 6 meters

AVIXA Domain: AV Design and Integration | Chapter Reference: Ch. 8-9

Unit 5 Quick-Check — Chapters 9-10: AV Integration & Maintenance/Troubleshooting

UNIT 5

Q117. After completing an AV system installation in a new school media center, the integrator provides the client with a package of documents. Which document set MOST completely represents a professional handover package?

- A. Only the purchase receipts for all equipment
- B. As-built drawings, equipment manuals, warranty documentation, control system source code, cable schedules, and a training record
- C. The original design proposal only
- D. A single-page system diagram with no cable labels

AVIXA Domain: AV Design and Integration | Chapter Reference: Ch. 9

Q118. A teacher reports that the classroom display turns on correctly but shows "No Signal" when her laptop is connected via HDMI. She has tried a different HDMI cable with the same result. What should be checked NEXT?

- A. Whether the display's audio output is muted
- B. Whether the laptop is outputting a video signal to the HDMI port — check display output settings (extend, mirror, or second-screen mode) on the laptop
- C. Whether the projector lamp is past its rated hours
- D. Whether the network switch has an IP address conflict

AVIXA Domain: Maintenance and Troubleshooting | Chapter Reference: Ch. 10

Q119. A school district implements a preventive maintenance schedule for all classroom AV equipment. Which of the following tasks is MOST appropriately scheduled on a semi-annual (every 6 months) basis?

- A. Replacing all display units
- B. Cleaning projector air filters, checking and logging lamp hours, inspecting all cable connections for secure seating, and testing wireless microphone battery life
- C. Replacing all wireless microphone transmitters
- D. Reprogramming all control system touch panels

AVIXA Domain: Maintenance and Troubleshooting | Chapter Reference: Ch. 10

Q120. A teacher using a document camera notices the image on the display is upside down. This is MOST easily corrected by:

- A. Replacing the document camera with a new unit
- B. Rotating the displayed image 180 degrees using the document camera's built-in rotation setting or software control
- C. Reinstalling the control system program
- D. Adjusting the projector's keystone correction vertically

AVIXA Domain: Maintenance and Troubleshooting | Chapter Reference: Ch. 10

Section 3: Complete Answer Key with Rationales (Q1-Q120)

How to Use This Answer Key

Read the rationale for every question — not just the ones you missed. Understanding *why* each distractor is incorrect is as educationally valuable as knowing the correct answer. Pay attention to the Bloom's Level tag for each question to ensure you are studying at the right cognitive depth for exam readiness.

Domain 1: AV Industry Overview & Fundamentals

Q1: **Correct Answer: B — AVIXA**

Rationale: AVIXA (Audiovisual and Integrated Experience Association) is the global trade association that develops and publishes technical standards for the AV industry, including AVIXA standards for display image size, audio coverage, and more. IEEE develops broader electronics and networking standards; ANSI oversees standards accreditation in the U.S.; TIA focuses on telecommunications infrastructure standards. While all four organizations produce relevant standards, AVIXA is the primary body for AV-specific standards.

Q2: Correct Answer: B — Combining audio, video, and control technologies into a unified, functional system

Rationale: AV integration means designing and installing audio, video, and control systems so they work together seamlessly within a space. Option A is incorrect — AV integration enhances teacher capability, not replaces it. Option C describes a fragmented, non-integrated approach. Option D describes internet streaming, which may be a component of an AV system but does not define integration itself.

Q3: Correct Answer: B — A representation of information carried by an electrical or optical medium

Rationale: In AV, a signal is a form of energy (electrical, optical, or electromagnetic) that carries audio, video, or control information from one device to another. Option A describes electrical power, not a signal. Option C is too narrow (signals include audio, data, and control, not only video, and are not exclusively wireless). Option D describes a connector, not the signal itself.

Q4: Correct Answer: B — AV technologist / systems technician

Rationale: The AV technologist (AVT) is the entry-level AV professional responsible for operating, supporting, and maintaining installed AV systems. The AV designer creates the design documents; the salesperson facilitates equipment procurement; the electrical contractor handles power infrastructure. After installation, the technologist ensures day-to-day operation aligns with design intent.

Q5: Correct Answer: C — Digital signals can be copied, processed, and transmitted with minimal degradation

Rationale: Digital signals represent information as discrete binary values, so they can be regenerated exactly during retransmission, preventing the cumulative noise and distortion that degrade analog signals. Option A is false — digital video typically requires more bandwidth than analog equivalents. Option B overstates immunity; digital signals can still be disrupted by extreme interference. Option D is false — digital signals still require amplification over long distances.

AVIXA Domain: AV Industry Overview & Fundamentals | ISTE Standard: N/A | *Bloom's Level: Understand*

Q6: Correct Answer: B — Assistive listening systems that transmit audio wirelessly to hearing-impaired students

Rationale: UDL's "multiple means of representation" principle requires information to be accessible in multiple formats. Assistive listening systems (FM, IR, or hearing loop/telecoil) provide audio directly to students with hearing loss, directly supporting this principle and ADA accessibility requirements. Options A, C, and D improve visual display quality or convenience but do not specifically address the multiple-means-of-representation principle for students with hearing differences.

AVIXA Domain: AV Industry Overview & Fundamentals | ISTE Standard: 2 – Leader | *Bloom's Level: Apply*

Q7: Correct Answer: B — They provide a framework for how educators and students should use technology to enhance learning

Rationale: ISTE Standards are a competency framework guiding how educators, students, and institutions should approach technology use for learning — they do not specify hardware, supersede district policy, or carry force of federal law. Options A, C, and D mischaracterize the nature and authority of the ISTE Standards.

AVIXA Domain: AV Industry Overview & Fundamentals | ISTE Standard: All | *Bloom's Level: Remember*

Q8: Correct Answer: B — A scope of work / needs assessment

Rationale: The needs assessment (or scope of work document) is the foundational document that captures user requirements, functional goals, constraints, and budget parameters before any design or procurement begins. A purchase order follows equipment selection; a cable schedule is a design output; as-built drawings document what was actually installed. All come later in the project lifecycle.

AVIXA Domain: AV Industry Overview & Fundamentals | ISTE Standard: 5 – Designer | *Bloom's Level: Apply*

Q9: Correct Answer: B — Distributing audio and video signals across a standard network infrastructure using IP protocols

Rationale: AV over IP (AVoIP) routes AV signals as data packets over standard IP networks (typically Ethernet), replacing dedicated point-to-point AV cables. Option A confuses AVoIP with consumer IoT devices. Option C describes internet streaming, which is a specific application of IP technology but not the definition of AVoIP in the AV infrastructure sense. Option D describes control-system remote operation, not AVoIP signal distribution.

AVIXA Domain: AV Industry Overview & Fundamentals | ISTE Standard: N/A | *Bloom's Level: Remember*

Q10: Correct Answer: B — Software-defined AV over IP with centralized management

Rationale: Software-defined AVoIP platforms allow centrally managed, scalable distribution of AV content across campus networks, adapting to changing needs without rewiring. VHS, coaxial analog switchers, and stand-alone Blu-ray players represent legacy or declining technologies that lack the scalability, flexibility, and remote management capabilities needed for modern campus-wide AV distribution.

AVIXA Domain: AV Industry Overview & Fundamentals | ISTE Standard: N/A | *Bloom's Level: Analyze*

Domain 2: Audio Systems

Q11: Correct Answer: B — Classroom audio distribution system with ceiling-mounted speakers

Rationale: When students in the back of a room cannot hear well, the root cause is insufficient sound level at their seating positions — a classic audio distribution problem. A

classroom audio system with properly spaced ceiling speakers brings the sound source closer to all listeners, dramatically improving clarity and reducing the teacher's vocal strain. Display improvements (Options A, C, D) address visibility, not audibility.

AVIXA Domain: Audio Systems | ISTE Standard: 6 – Facilitator | *Bloom's Level: Apply*

Q12: Correct Answer: B — A logarithmic ratio comparing a measured level to a reference level

Rationale: The decibel is a dimensionless logarithmic ratio — it expresses how much larger or smaller a measured quantity is compared to a defined reference. Option A incorrectly describes an absolute pressure measurement (the Pascal is the unit of sound pressure, and 20 μPa is the threshold of hearing, not 1 Pa). Options C and D describe different quantities (frequency and impedance) measured in different units.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | *Bloom's Level: Remember*

Q13: Correct Answer: C — 10 dBu

Rationale: Decibels express gain/loss as a ratio. If the input is 0 dBu (the reference level) and the amplifier adds 10 dB of gain, the output is $0 + 10 = 10$ dBu. This is a straightforward application of the additive nature of decibel values. Option A would indicate no gain. Option B would indicate only 5 dB gain. Option D (100 dBu) would require a gain of 100 dB, which would be extraordinarily large.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | *Bloom's Level: Apply*

Q14: Correct Answer: B — Dynamic microphone

Rationale: A dynamic (moving-coil) microphone generates voltage through electromagnetic induction: a coil attached to the diaphragm moves within a magnetic field when sound waves cause the diaphragm to vibrate. A condenser microphone uses a capacitor principle requiring external power. A ribbon microphone also uses electromagnetic induction but with a thin metallic ribbon rather than a coil. A boundary microphone uses a pressure-zone design and is typically a condenser type.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | *Bloom's Level: Remember*

Q15: Correct Answer: C — 48V DC

Rationale: Standard phantom power (P48) supplies 48 volts DC through the XLR cable's pins 2 and 3 equally (relative to pin 1) to power condenser microphone circuitry. Lower voltages (9V, 24V) are used in some reduced-phantom-power systems for specific microphone designs. 120V AC is the wall outlet voltage, not a microphone power standard — applying it to a microphone would cause immediate damage.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | Bloom's Level: Remember

Q16: Correct Answer: C — The front of the microphone, with rejection at the rear

Rationale: A cardioid polar pattern is heart-shaped, picking up sound predominantly from the front (0°) and attenuating sound from the rear (180°). This makes it ideal for live stage use and classroom applications where rear rejection reduces feedback. Option A describes omnidirectional. Option B describes bidirectional (figure-eight). Option D partially describes supercardioid — a cardioid is broader in its front pickup than supercardioid.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | Bloom's Level: Remember

Q17: Correct Answer: B — Acoustic feedback caused by the microphone picking up its own amplified output

Rationale: Audio feedback (the sustained tone or squeal) occurs when the microphone picks up sound from the speakers, re-amplifies it, and feeds it back into the speakers in a continuous loop. Low microphone battery (A) would cause intermittent signal dropouts. A broken XLR connector (C) would cause signal loss, not a sustained tone. Impedance mismatch (D) causes level and tone coloration, not a feedback loop.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | Bloom's Level: Analyze

Q18: Correct Answer: C — Ohms (Ω)

Rationale: Impedance, the opposition to alternating current in an audio circuit, is measured in ohms (Ω). Decibels measure signal level ratios; Hertz measures frequency; Watts measure power. Understanding impedance is critical when matching microphones to mixers, mixers to amplifiers, and amplifiers to speakers to ensure maximum power transfer and avoid damage.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | Bloom's Level: Remember

Q19: Correct Answer: A — 20 Hz to 20,000 Hz

Rationale: The standard accepted range of human hearing is approximately 20 Hz (low bass) to 20,000 Hz or 20 kHz (high treble), though individual sensitivity varies with age and health. Option B (200–2,000 Hz) covers only the speech intelligibility range. Option C (100–10,000 Hz) partially overlaps but omits the lowest and highest extremes. Option D (50–50,000 Hz) extends beyond the human range into ultrasonic territory.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | *Bloom's Level: Remember*

Q20: Correct Answer: A — FM assistive listening system

Rationale: FM assistive listening systems (and also hearing loop/telecoil systems) are the most commonly deployed ADA-compliant assistive listening solutions in educational facilities. FM systems transmit audio on FM radio frequencies to personal receivers. AM broadcast is not designed for assistive listening applications. Bluetooth's latency, pairing complexity, and limited range make it less suitable for large-venue compliance. Infrared illuminators are used for IR remote control signals, not audio transmission (IR-based assistive listening is also used but less prevalently than FM in educational settings).

AVIXA Domain: Audio Systems | ISTE Standard: 2 – Leader | *Bloom's Level: Apply*

Q21: Correct Answer: B — Dispersion angle (coverage angle)

Rationale: To achieve even coverage across the audience, the designer must select speakers whose dispersion angle (typically specified as horizontal × vertical degrees at –6 dB) can cover all listener positions when mounted at the installation height. Maximum SPL (A) determines loudness but not spatial coverage. Input impedance (C) affects the electrical matching between amplifier and speaker. Subwoofer frequency response (D) is relevant to bass reproduction, not even midrange/treble coverage across the audience.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | *Bloom's Level: Apply*

Q22: Correct Answer: B — It provides common-mode noise rejection, reducing hum and interference over long cable runs

Rationale: Balanced connections use a differential signaling method: the same audio signal is sent in phase on one conductor and out-of-phase on another. Any noise picked up by the cable is identical on both conductors (common-mode), and the balanced receiver

rejects this common-mode noise, leaving only the desired audio. This makes balanced audio essential for long cable runs in noisy environments. Options A, C, and D misstate technical facts about balanced audio.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | Bloom's Level: Understand

Q23: Correct Answer: B — Manipulate audio signals using algorithms for equalization, dynamics, delay, and routing

Rationale: A DSP (Digital Signal Processor) is a programmable processor that performs mathematical operations on digitized audio signals to achieve equalization, compression, limiting, delay, room correction, echo cancellation, and signal routing. Option A describes a power supply. Option C describes fiber-optic transceivers. Option D describes wireless networking components.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | Bloom's Level: Remember

Q24: Correct Answer: B — A low-cut (high-pass) filter on the microphone channel

Rationale: "Boomy" or excessive bass on a spoken-voice microphone signal is typically caused by proximity effect (common with directional mics) or HVAC rumble. A low-cut (high-pass) filter rolls off frequencies below a set threshold (e.g., 80–120 Hz), removing the problematic bass buildup while leaving speech fundamentals and harmonics intact. A gate (A) controls noise floor, not tonal balance. A compressor (C) controls dynamic range, not frequency content. A delay line (D) addresses timing alignment, not tonal coloration.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | Bloom's Level: Apply

Q25: Correct Answer: B — Speaker spacing based on coverage angle and mounting height, using the tangent function

Rationale: Optimal ceiling speaker placement uses trigonometry: the radius of coverage at floor level is calculated as mounting height $\times \tan(\text{half the coverage angle})$. Spacing is then set so adjacent speaker coverage patterns overlap appropriately (typically at -6 dB points) to ensure even sound distribution. Option A is a gross oversimplification. Option C uses non-acoustic variables. Option D conflates electrical and acoustic calculations.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | Bloom's Level: Analyze

Domain 3: Video and Display Systems

Q26: Correct Answer: B — Lumens (brightness) output

Rationale: A washed-out, faded image in ambient light is a brightness problem — the projector's light output is insufficient to overcome the ambient light in the room. Lumens (ANSI lumens) is the specification that directly quantifies projector brightness. A higher lumen rating is needed for rooms that cannot be fully darkened. Contrast ratio (A) affects black levels but cannot compensate for insufficient brightness. Resolution (C) affects image sharpness, not brightness. Aspect ratio (D) affects image shape, not luminance.

AVIXA Domain: Video and Display Systems | ISTE Standard: 5 – Designer | Bloom's Level: Apply

Q27: Correct Answer: B — Throw ratio = Throw distance ÷ Screen width

Rationale: Throw ratio (TR) = Throw Distance / Image Width. A TR of 1.0 means the projector must be positioned at a distance equal to the screen width. A TR of 0.5 (short-throw) means the projector can be positioned at half the screen width. Option A inverts the formula. Options C and D mix in unrelated specifications (lumens, resolution, pixel density).

AVIXA Domain: Video and Display Systems | ISTE Standard: N/A | Bloom's Level: Remember

Q28: Correct Answer: C — Short-throw or ultra-short-throw projector

Rationale: Short-throw projectors have a low throw ratio (typically 0.4–0.8) and ultra-short-throw projectors have ratios below 0.4, allowing them to project a large image from a very short distance from the screen. This is ideal for space-constrained environments like libraries with low ceilings or limited throw depth. Standard-throw projectors (A) require greater distance. Long-throw projectors (B) require even greater distance. Rear projection (D) requires space behind the screen, which is even more space-intensive.

AVIXA Domain: Video and Display Systems | ISTE Standard: N/A | Bloom's Level: Apply

Q29: Correct Answer: B — The fixed number of physical pixels in the display panel

Rationale: Native resolution refers to the actual, fixed pixel grid manufactured into the display panel. When content matches the native resolution, each source pixel corresponds

exactly to one display pixel, producing the sharpest image. Option A describes the maximum accepted input resolution, which may be higher than native. Option C confuses resolution with color calibration settings. Option D describes a legacy compatibility mode, not native resolution.

AVIXA Domain: Video and Display Systems | ISTE Standard: N/A | *Bloom's Level: Remember*

Q30: Correct Answer: C — 3840 × 2160

Rationale: 4K UHD (Ultra High Definition) has a resolution of 3840 × 2160 pixels — exactly four times the pixel count of 1080p Full HD. Option A (1920×1080) is 1080p Full HD. Option B (2560×1440) is 2K QHD. Option D (7680×4320) is 8K UHD. Understanding resolution standards is fundamental for AV practitioners specifying and installing displays.

AVIXA Domain: Video and Display Systems | ISTE Standard: N/A | *Bloom's Level: Remember*

Q31: Correct Answer: C — Superior brightness and contrast in ambient light conditions

Rationale: Large-format LCD flat panels emit light directly and can achieve very high brightness (often 400–700 nits or more), making them far more readable in lit rooms than projected images. Option A is often incorrect — flat panels tend to have higher upfront cost, though total cost may vary. Option B is incorrect — projectors can produce larger images than most fixed panels. Option D is incorrect — large flat panels are not portable.

AVIXA Domain: Video and Display Systems | ISTE Standard: N/A | *Bloom's Level: Analyze*

Q32: Correct Answer: A — The ratio of maximum brightness to minimum brightness (black level)

Rationale: Contrast ratio is expressed as the ratio of the display's brightest white to its darkest black (e.g., 1000:1). A higher contrast ratio generally indicates a display can produce more vivid, detailed images with better shadow detail and perceived color depth. Option B describes aspect ratio. Option C describes color gamut or color temperature range. Option D describes refresh rate differences between standards.

AVIXA Domain: Video and Display Systems | ISTE Standard: N/A | *Bloom's Level: Remember*

Q33: Correct Answer: B — Minimum contrast ratio of 4.5:1 for normal-sized text

Rationale: WCAG 2.1 Level AA requires a minimum contrast ratio of 4.5:1 between text and its background for normal-size text (and 3:1 for large text, 18pt+ or 14pt bold). Option A (light gray on white) would likely fail this threshold. Option C (2-second cycling) would violate WCAG 2.3.1 (seizures) and 2.2.2 (auto-updating content). Option D (animated GIFs without captions) would violate 1.2.1 (captions for multimedia).

AVIXA Domain: Video and Display Systems | ISTE Standard: 2 – Leader | *Bloom's Level: Apply*

Q34: Correct Answer: B — HD and UHD video content and modern widescreen displays

Rationale: The 16:9 aspect ratio is the established standard for HDTV, Full HD (1080p), 4K UHD, and virtually all modern widescreen displays and presentations. Option A (35mm film) uses a 1.37:1 or 1.85:1 aspect ratio. Option C (standard-definition TV) used 4:3. Option D (DCI scope) uses 2.39:1 for cinematic widescreen.

AVIXA Domain: Video and Display Systems | ISTE Standard: N/A | *Bloom's Level: Remember*

Q35: Correct Answer: B — Direct-view LED uses individual LED modules as pixels rather than a backlit LCD panel

Rationale: In a direct-view LED display (also called fine-pitch LED or dvLED), each pixel consists of individual red, green, and blue LED elements. Panels can be tiled seamlessly for any size, achieving very high brightness and wide color gamut. Option A is false — LEDs reproduce color very accurately. Option C is false — direct-view LED accepts standard digital inputs. Option D is false — large LED walls typically consume more power than a single equivalent LCD panel.

AVIXA Domain: Video and Display Systems | ISTE Standard: N/A | *Bloom's Level: Understand*

Q36: Correct Answer: B — Source resolution mismatch causes image scaling artifacts

Rationale: When a source outputs 1920×1080 to a projector with a native resolution of 1024×768, the projector must scale (downconvert) the image, which introduces interpolation artifacts that appear as blurriness, jagged edges, or moiré patterns on text. The correct fix is to match the source output resolution to the projector's native resolution,

or upgrade the projector. Lamp life (A) causes brightness reduction, not fuzziness. Cable length (C) affects signal integrity differently. Contrast ratio (D) affects black levels, not sharpness.

AVIXA Domain: Video and Display Systems | ISTE Standard: N/A | *Bloom's Level: Analyze*

Q37: Correct Answer: B — A large-format display and a touch-sensitive input layer

Rationale: An interactive flat-panel display (IFPD) integrates a large LCD (or LED) display with an infrared or capacitive touch layer, allowing users to write, draw, annotate, and interact directly on the screen surface. Options A, C, and D describe other individual AV components or partial combinations that do not constitute an IFPD.

AVIXA Domain: Video and Display Systems | ISTE Standard: 6 – Facilitator | *Bloom's Level: Remember*

Q38: Correct Answer: C — 6× screen height

Rationale: AVIXA/ANSI display standards specify that the maximum viewing distance for standard-definition content is approximately 6 times the screen height (6H). For HD content, the maximum extends to approximately 8H, and the minimum viewing distance is 1.5H. Option A (2H) is well within the minimum viewing distance. Option B (4H) falls between minimum and maximum. Option D (8H) applies to HD content, not standard definition.

AVIXA Domain: Video and Display Systems | ISTE Standard: N/A | *Bloom's Level: Remember*

Q39: Correct Answer: B — A digital signage media player connected to a content management system (CMS)

Rationale: Dynamic, remotely updated digital signage requires a media player (a dedicated computing device) connected to a CMS (software platform) that pushes updated content to the player on a schedule or in real time. A Blu-ray player (A) can only play pre-recorded discs. A document camera (C) captures live objects, not managed content. An analog cable TV tuner (D) receives broadcast TV, not dynamic digital content.

AVIXA Domain: Video and Display Systems | ISTE Standard: N/A | *Bloom's Level: Apply*

Q40: Correct Answer: B — A digital or optical adjustment that corrects trapezoidal image distortion when the projector is not perpendicular to the screen

Rationale: Keystone distortion appears as a trapezoidal (wedge-shaped) image when the projector is angled up, down, or sideways relative to the screen. Keystone correction — available digitally in the projector's image processing — adjusts the image geometry to produce a rectangular output. Option A describes color calibration. Option C describes a lens shading or gain feature. Option D describes resolution/format matching.

AVIXA Domain: Video and Display Systems | ISTE Standard: N/A | *Bloom's Level: Understand*

Domain 4: Signal Flow and Connectivity

Q41: Correct Answer: B — The path audio and/or video signals travel from source through processing and distribution to an output

Rationale: Signal flow is the conceptual model describing how AV signals originate at a source (camera, laptop, microphone), are processed/switched, and ultimately reach an output (display, speaker). It is the fundamental framework for designing, troubleshooting, and documenting AV systems. Option A describes power distribution. Option C describes physical cable routing. Option D describes power sequencing.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | *Bloom's Level: Remember*

Q42: Correct Answer: C — HDMI

Rationale: HDMI (High-Definition Multimedia Interface) carries uncompressed digital video and multi-channel digital audio on a single cable and is the predominant connection between laptops and classroom displays. VGA (A) carries analog video only (no audio). XLR (B) is a professional audio connector. RS-232 (D) is a serial control protocol, not an AV signal carrier.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | *Bloom's Level: Remember*

Q43: Correct Answer: B — DisplayPort uses a packet-based protocol better for high-refresh-rate and multi-stream; HDMI is more prevalent in consumer electronics

Rationale: Both HDMI and DisplayPort are digital. DisplayPort's packet-based architecture supports higher refresh rates, adaptive sync, and Multi-Stream Transport (daisy-chaining multiple monitors). HDMI is more common in consumer electronics (TVs, projectors). Option A is false — both are digital. Option C is false — DisplayPort does carry audio. Option D is context-dependent and not universally true; HDMI 2.1 supports 10K resolution, matching or exceeding DisplayPort 2.0 in some configurations.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | Bloom's Level: Understand

Q44: Correct Answer: B — USB-C to HDMI adapter (using USB-C with DisplayPort Alt Mode or Thunderbolt)

Rationale: USB-C ports that support DisplayPort Alt Mode or Thunderbolt 3/4 can output video signals through a USB-C to HDMI adapter, allowing direct connection to HDMI displays. USB-C to VGA (A) would produce an analog video signal that many modern displays may not accept. USB-C to RS-232 (C) is a serial control adapter, not a video output. USB-C to XLR (D) does not exist as a standard AV product.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | Bloom's Level: Apply

Q45: Correct Answer: C — Fiber-optic HDMI or HDBaseT over fiber

Rationale: Standard passive HDMI copper cables are limited to approximately 5–15 meters for 4K signals. For 100-meter runs, fiber-optic HDMI cables or HDBaseT-over-fiber extenders are required. These use optical signals immune to electromagnetic interference and capable of very long runs without signal degradation. Composite video (B) is a low-quality analog standard. Cat 5e without an extender (D) can carry HDBaseT only up to 100m when specifically designed for that protocol with an active transmitter/receiver.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | Bloom's Level: Apply

Q46: Correct Answer: B — Uncompressed HD/4K video, audio, control, USB, and PoE over a single Cat 5e/6 cable

Rationale: HDBaseT is a connectivity standard that transmits all five elements — HD or 4K video, multi-channel audio, RS-232 and IR control, USB 2.0, and PoE — over a single Cat 5e or Cat 6 cable up to 100 meters. This is its key advantage over HDMI alone. Option A is incomplete. Option C incorrectly states the signal is compressed. Option D is incomplete and states an incorrect distance limit.

Q47: Correct Answer: B — HDMI cable not rated for 4K bandwidth or exceeding passive cable distance limits

Rationale: HDMI 2.0 (required for 4K/60Hz) must support 18 Gbps. Standard HDMI cables for this bandwidth are rated for approximately 3–5 meters passively. Longer runs require active cables, repeaters, or extenders. Without sufficient bandwidth, the HDMI handshake may fall back to a supported lower resolution (1080p). The display's capability (A) would produce no image at all if it were the issue. Source aspect ratio (C) and ARC (D) are unrelated to resolution fall-back.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | Bloom's Level: Analyze

Q48: Correct Answer: B — To route any input signal to any output destination independently and simultaneously

Rationale: A matrix switcher provides "any input to any output" routing — in an M×N matrix, any of M inputs can be connected to any combination of N outputs simultaneously and independently. This is essential for multi-room, multi-source AV systems. Option A describes a converter. Option C describes a distribution amplifier or signal repeater. Option D describes an audio mixer.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | Bloom's Level: Remember

Q49: Correct Answer: A — The projector is not HDCP-compliant and the source is transmitting protected content

Rationale: HDCP establishes an encrypted handshake between a source and a display. If any device in the signal chain (including switchers, extenders, or the projector) is not HDCP-compliant, the handshake fails and the source may output a blank screen or reduced resolution. The laptop's local monitor remains functional because it doesn't require HDCP. Cable length (B), brightness (C), and adapter count (D) are unrelated to HDCP authentication failures.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | Bloom's Level: Analyze

Q50: Correct Answer: C — XLR

Rationale: The XLR connector (most commonly a 3-pin version) is the standard for balanced professional audio connections: Pin 1 = ground/shield, Pin 2 = signal positive (hot), Pin 3 = signal negative (cold). RCA (A) is an unbalanced consumer connector. 3.5mm TRS (B) can carry balanced audio in some configurations but is a consumer/prosumer connector. TS (D) is a mono unbalanced phone connector.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | *Bloom's Level: Remember*

Q51: Correct Answer: B — Labeling at both ends of every cable with consistent identifiers matching as-built documentation

Rationale: Industry best practice requires labeling at both ends of every cable with a consistent identifier (typically alphanumeric) that matches the project's cable schedule and as-built drawings. This enables any technician to trace any signal path, troubleshoot, or make changes years after installation. Labeling only at origin (A) creates confusion at the destination end. Color-coding only (C) is insufficient for complex systems. Selective labeling (D) leaves unlabeled cables that are impossible to trace.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | *Bloom's Level: Apply*

Q52: Correct Answer: B — To split and amplify a single AV signal to feed multiple outputs simultaneously without signal loss

Rationale: A distribution amplifier (DA) takes a single AV input and creates multiple identical outputs, each with appropriate signal strength. Without a DA, splitting a signal passively (with a Y-cable or passive splitter) causes significant level loss due to impedance loading. A DA solves this by buffering and re-amplifying each output. Option A describes a converter/DAC. Option C describes a switcher or matrix. Option D describes a mixer preamp.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | *Bloom's Level: Remember*

Q53: Correct Answer: B — Insert a ground loop isolator or DI box with a ground-lift switch

Rationale: Ground loop hum (60 Hz in North America, 50 Hz in Europe) occurs when two devices connected together have different ground potentials, creating a current loop that induces the characteristic low hum. A ground loop isolator breaks the DC continuity (lifting the ground path) while maintaining the audio signal connection, eliminating the loop.

Increasing cable length (A) worsens the problem. Increasing gain (C) amplifies the hum along with the signal. Changing USB ports (D) does not affect the audio ground path.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | *Bloom's Level: Apply*

Q54: Correct Answer: B — Allow source to query the display's supported resolutions, refresh rates, and audio formats for automatic format negotiation

Rationale: EDID is a metadata standard embedded in HDMI and DisplayPort displays. When a source connects, it reads the display's EDID data block to discover what resolutions, refresh rates, color depths, and audio formats the display supports, then configures its output accordingly. HDCP (A) handles content encryption. PoE (C) and remote calibration (D) are not EDID functions.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | *Bloom's Level: Understand*

Q55: Correct Answer: B — Latency, compression artifacts, and network congestion may degrade the experience

Rationale: Wireless presentation systems compress and wirelessly transmit video, introducing latency (perceptible delay between presenter movement and display update) and compression artifacts. On shared school Wi-Fi, network congestion from concurrent student device use can exacerbate these issues. Option A is false — wired HDMI is generally superior. Option C is false — most modern wireless systems do carry audio. Option D is false — many systems support 4K at compressed quality.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | *Bloom's Level: Analyze*

Domain 5: AV Networking

Q56: Correct Answer: D — 3 Gbps or more

Rationale: Uncompressed 1080p60 video with full color (4:2:2 or 4:4:4) requires approximately 3 Gbps of raw bandwidth, which explains why uncompressed AVoIP requires 10 Gbps network infrastructure per stream. This is why most AVoIP systems use some level of compression (e.g., JPEG 2000, H.264, H.265) to reduce bandwidth to a manageable 1 Gbps or less per stream. Options A and B are far too low for uncompressed signals. Option C (1 Gbps) would support compressed but not fully uncompressed 1080p60.

Q57: Correct Answer: B — DHCP (Dynamic Host Configuration Protocol)

Rationale: DHCP automatically assigns IP addresses, subnet masks, default gateways, and DNS server addresses to devices on the network, eliminating the need to manually configure each AV endpoint. DNS (A) resolves hostnames to IP addresses. SNMP (C) is used for network device monitoring and management. NTP (D) synchronizes device clocks across the network.

AVIXA Domain: AV Networking | ISTE Standard: N/A | Bloom's Level: Remember

Q58: Correct Answer: B — Segmenting AV traffic onto a dedicated VLAN and applying QoS policies

Rationale: VLANs (Virtual LANs) logically separate network traffic on the same physical infrastructure, preventing AV streams from competing with student internet traffic for bandwidth. QoS policies further ensure AV traffic receives priority treatment. Option A is expensive and defeats the purpose of shared network infrastructure. Option C degrades AV quality unnecessarily. Option D is a step backward in capability.

AVIXA Domain: AV Networking | ISTE Standard: N/A | Bloom's Level: Apply

Q59: Correct Answer: B — The ability to prioritize specific types of traffic to ensure consistent performance and low latency

Rationale: QoS (Quality of Service) is a set of network techniques that prioritize certain traffic types (audio/video) over others (general data downloads) to ensure time-sensitive AV streams receive guaranteed bandwidth and low, consistent latency. It is implemented through traffic shaping, packet marking (DSCP), and priority queuing. Option A describes cable quality standards. Option C describes network security (WPA2/3). Option D describes switch port capacity.

AVIXA Domain: AV Networking | ISTE Standard: N/A | Bloom's Level: Understand

Q60: Correct Answer: B — 192.168.0.0 - 192.168.255.255

Rationale: RFC 1918 defines three private IPv4 address ranges not routable on the public internet: 10.0.0.0-10.255.255.255, 172.16.0.0-172.31.255.255, and 192.168.0.0-

192.168.255.255. Option B is one of these. Options A, C, and D are not RFC 1918 private ranges. Understanding private addressing is fundamental for configuring AV devices on school networks.

AVIXA Domain: AV Networking | ISTE Standard: N/A | *Bloom's Level: Remember*

Q61: Correct Answer: B — Multicast sends a single stream that multiple receivers join, reducing bandwidth vs. separate unicast streams

Rationale: In unicast, the server sends a separate copy of the stream to each receiver — with 20 receivers, bandwidth is multiplied by 20. In multicast, the server sends one stream; the network delivers it only to devices that have joined the multicast group, scaling efficiently regardless of receiver count. Option A is false — multicast works on standard Ethernet. Option C is false — multicast requires managed switches for proper control. Option D is false — codec selection is independent of multicast/unicast.

AVIXA Domain: AV Networking | ISTE Standard: N/A | *Bloom's Level: Understand*

Q62: Correct Answer: B — Managed switches support VLAN, QoS, IGMP snooping, and traffic monitoring

Rationale: AV-over-IP systems require managed switches because they provide the configuration capability for VLANs (traffic segmentation), QoS (traffic prioritization), and IGMP snooping (multicast management) that unmanaged switches lack. Options A, C, and D are incorrect — unmanaged switches can forward Ethernet frames, and switches do not amplify AV signals or convert IP to HDMI.

AVIXA Domain: AV Networking | ISTE Standard: N/A | *Bloom's Level: Understand*

Q63: Correct Answer: B — Changing default credentials, applying firmware updates, and placing AV devices on isolated segments

Rationale: AV devices shipped with default admin credentials (e.g., "admin/admin") are a known cybersecurity vulnerability. Best practice requires immediately changing credentials, applying current firmware to patch known vulnerabilities, and isolating AV devices on a separate network segment or VLAN. Option A (using default credentials) is the most common AV cybersecurity vulnerability exploited. Options C and D create additional attack surface.

AVIXA Domain: AV Networking | ISTE Standard: 3 – Citizen | *Bloom's Level: Apply*

Q64: Correct Answer: B — Allows the switch to forward multicast streams only to ports where receivers have subscribed

Rationale: Without IGMP snooping, a managed switch treats multicast traffic like broadcast traffic and floods it to all ports, consuming bandwidth on all connected devices. IGMP snooping allows the switch to inspect IGMP join/leave messages and maintain a table of which ports have active multicast subscribers, forwarding streams only to those ports. Option A describes encryption. Option C describes transcoding. Option D describes jitter buffering.

AVIXA Domain: AV Networking | ISTE Standard: N/A | *Bloom's Level: Understand*

Q65: Correct Answer: A — Verify all AV endpoints have static or reserved IP addresses with no IP conflicts

Rationale: After confirming the switch and QoS are properly configured, the next step in a systematic troubleshooting approach is to verify IP addressing. AV-over-IP endpoints require stable, unique IP addresses — DHCP lease renewals or duplicate IP addresses can cause intermittent disconnections. Options B, C, and D address display, audio, and acoustic issues that are unrelated to intermittent network video stream drops.

AVIXA Domain: AV Networking | ISTE Standard: N/A | *Bloom's Level: Analyze*

Domain 6: Control Systems

Q66: Correct Answer: B — Providing a centralized, programmable interface to operate multiple AV devices through a single interface

Rationale: A control system integrates diverse AV devices (projectors, displays, amplifiers, screens, lighting, shading) into a unified, user-operable system accessible through a touch panel, button panel, or mobile device. This simplifies AV operation for non-technical users. Options A, C, and D describe functions of audio amplifiers, video converters, and PoE switches, respectively.

AVIXA Domain: Control Systems | ISTE Standard: N/A | *Bloom's Level: Remember*

Q67: Correct Answer: B — RS-232 serial communication

Rationale: RS-232 is a serial communication standard widely used by AV control systems to send commands (such as power on/off, input selection, volume) to AV equipment via a dedicated serial port. Most professional projectors and displays include an RS-232 port with a documented command set. HDMI CEC (A) is a limited consumer-focused control protocol. DMX-512 (C) is used for theatrical lighting control. MIDI (D) is a musical instrument control protocol.

AVIXA Domain: Control Systems | ISTE Standard: N/A | *Bloom's Level: Remember*

Q68: Correct Answer: C — Simplifying the interface to display only the most-used functions with clear icons and labels

Rationale: Good AV UI design follows human factors principles: expose only the functions users actually need, use clear icons and plain-language labels, and design for the least-experienced anticipated user (substitute teachers, visiting staff). Option A (more buttons) increases cognitive load and error rates. Option B (small text) reduces readability. Option D (command line) requires technical expertise incompatible with the user population.

AVIXA Domain: Control Systems | ISTE Standard: 6 – Facilitator | *Bloom's Level: Apply*

Q69: Correct Answer: A — Sending encoded light pulses in the IR spectrum from a blaster to a device's IR receiver

Rationale: IR control uses modulated infrared light pulses (typically 30–60 kHz carrier frequency) to transmit encoded command signals to a device's IR receiver — replicating remote control functionality. IR requires line-of-sight and cannot penetrate walls or work in bright IR ambient light without IR blasters. Option B describes RF (radio frequency) control. Option C describes IP-based network control. Option D describes RS-485 serial communication.

AVIXA Domain: Control Systems | ISTE Standard: N/A | *Bloom's Level: Remember*

Q70: Correct Answer: B — Baud rate, data bits, stop bits, and parity settings must match between the control processor and display

Rationale: RS-232 serial communication requires exact parameter matching between sender and receiver: baud rate (communication speed), data bits (typically 8), stop bits (1 or 2), and parity (none, even, odd). A mismatch causes garbled or ignored commands even

with a perfect physical connection. Display input selection (A), lamp hours (C), and DHCP leases (D) are unrelated to RS-232 command delivery.

AVIXA Domain: Control Systems | ISTE Standard: N/A | *Bloom's Level: Analyze*

Q71: Correct Answer: B — Occupancy sensor integration with control system logic

Rationale: Occupancy sensors (passive infrared or ultrasonic) detect human presence in a room. When integrated with the AV control system, the processor can trigger a power-down sequence after a configurable inactivity period, reducing energy consumption when rooms are unoccupied. This is a significant contributor to a school's sustainability goals. Options A, C, and D address display scaling, video format negotiation, and network traffic — not energy management.

AVIXA Domain: Control Systems | ISTE Standard: N/A | *Bloom's Level: Apply*

Q72: Correct Answer: B — Communicate with external software platforms through standardized data exchange

Rationale: An API (Application Programming Interface) defines a standardized way for software systems to exchange data and commands. An AV control system with API support can integrate with room booking systems, calendar platforms, building management systems, and UC platforms — enabling automated, data-driven AV workflows. Options A, C, and D describe physical cable connections, audio format conversion, and signal bandwidth, respectively — none of which are API functions.

AVIXA Domain: Control Systems | ISTE Standard: N/A | *Bloom's Level: Understand*

Q73: Correct Answer: A — A preset macro that executes a programmed sequence of commands automatically

Rationale: A macro is a pre-programmed sequence of control commands triggered by a single user action (button press). For a "class start" macro: press one button → display powers on (RS-232) → input switches to laptop (matrix command) → volume set to preset level (DSP command) → screen lowers (relay). Option B (individual IR remotes) requires multiple separate actions. Option C (relay) can only toggle power circuits, not execute multi-device sequences. Option D (OSD menu) requires direct access to each device.

AVIXA Domain: Control Systems | ISTE Standard: 6 – Facilitator | *Bloom's Level: Apply*

Q74: Correct Answer: B — Unauthorized access could allow an intruder to manipulate AV devices, disable displays, or use the system as a network entry point

Rationale: IP-connected control processors are network-attached computing devices. If inadequately secured, they can be accessed by unauthorized parties who could disrupt school AV operations, use the device as a pivot point to access other school network resources, or conduct denial-of-service attacks. Options A, C, and D misidentify network connectivity as causing audio or electrical issues, which is incorrect.

AVIXA Domain: Control Systems | ISTE Standard: 3 – Citizen | *Bloom's Level: Analyze*

Q75: Correct Answer: B — Environmental integration or "smart room" automation

Rationale: When an AV control system coordinates AV equipment with the room environment — automated shading, lighting scenes, HVAC setpoints, door locks — the result is called environmental integration or smart room automation. This extends the control system beyond pure AV functions into a full room experience layer. Options A, C, and D refer to unrelated AV technical concepts.

AVIXA Domain: Control Systems | ISTE Standard: N/A | *Bloom's Level: Understand*

Domain 7: Collaboration and UC Systems

Q76: Correct Answer: B — The room microphone is not routed to the video conferencing system's audio input

Rationale: For remote participants to hear in-room students, the room's microphone system must be connected to (and routed through) the video conferencing system's audio input. This is a common integration gap in hybrid classroom setups. Display resolution (A), HDMI connectivity (C), and internet speed (D) would not cause an audio routing problem between the in-room microphones and the UC system.

AVIXA Domain: Collaboration and UC Systems | ISTE Standard: 6 – Facilitator | *Bloom's Level: Analyze*

Q77: Correct Answer: B — Designing AV and UC systems so remote participants have an equivalent learning experience to in-room participants

Rationale: Hybrid equity is an instructional design and AV design principle that demands remote learners not be disadvantaged compared to in-room participants. This includes audio clarity (hearing all in-room voices), visual access (seeing the teacher, whiteboard, and fellow students), and interactive participation capability (being heard, seen, and engaged). Option A misinterprets equity as a numbers issue. Options C and D address components but not the overarching equity principle.

AVIXA Domain: Collaboration and UC Systems | ISTE Standard: 2 – Leader | *Bloom's Level: Understand*

Q78: Correct Answer: B — A hardware or software system that compresses (encodes) and decompresses (decodes) audio and video signals for transmission

Rationale: "Codec" is a portmanteau of "coder-decoder." In video conferencing, codecs (such as H.264, H.265/HEVC, VP8, VP9, AV1) compress audio and video for efficient network transmission and decompress them for playback. Option A describes a cable. Option C describes a remote control. Option D describes a power supply unit.

AVIXA Domain: Collaboration and UC Systems | ISTE Standard: N/A | *Bloom's Level: Remember*

Q79: Correct Answer: B — Auto-framing / presenter tracking camera with pan, tilt, and zoom capability

Rationale: A presenter-tracking PTZ camera automatically detects and follows the teacher as she moves around the room using computer vision, keeping her centered and in frame. This provides the best remote participant experience in a dynamic classroom. A fixed wide-angle lens (A) would make the teacher appear small and may not clearly show facial expressions. A document camera (C) captures desk-level objects, not the teacher. A laptop webcam (D) would not track a mobile presenter.

AVIXA Domain: Collaboration and UC Systems | ISTE Standard: 6 – Facilitator | *Bloom's Level: Apply*

Q80: Correct Answer: B — The far-end participants' audio being picked up by the room microphone and fed back to them as an echo

Rationale: AEC (Acoustic Echo Cancellation) prevents the audio from the far-end (the remote participants' voices played through the room speaker) from being picked up by the room microphone and retransmitted back to those same participants as a delayed echo. Without AEC, far-end participants hear their own voices echoed back, making conversation impossible. Options A, C, and D describe HVAC noise, display distortion, and network issues — not echo.

AVIXA Domain: Collaboration and UC Systems | ISTE Standard: N/A | *Bloom's Level: Understand*

Q81: Correct Answer: B — Whether the platform supports standard protocols (SIP/H.323) or has certified hardware integrations with the installed AV equipment

Rationale: Hardware interoperability is the critical technical consideration when selecting a UC platform for a hardware-equipped room AV system. Platforms that support open standards (SIP, H.323) or provide certified integrations with room AV hardware ensure reliable, full-featured operation. Cost, social features, and analog-only audio are secondary or incorrect considerations for an AV-integrated environment.

AVIXA Domain: Collaboration and UC Systems | ISTE Standard: N/A | *Bloom's Level: Apply*

Q82: Correct Answer: B — Voice activity detection (VAD) with single-talker priority

Rationale: Many AEC/DSP systems use voice activity detection to identify when someone is speaking and activate the microphone accordingly. In some single-talker priority configurations, only the loudest or first-detected voice is transmitted, causing other simultaneous speakers to be dropped. This can be mitigated by configuring the DSP for multi-talker mixing. AGC (A) adjusts level but doesn't mute simultaneous voices. Acoustic isolation (C) is a physical treatment. Noise gate (D) would cut low-level signals, but the described issue is between simultaneous talkers.

AVIXA Domain: Collaboration and UC Systems | ISTE Standard: N/A | *Bloom's Level: Analyze*

Q83: Correct Answer: B — To allow teachers to share physical documents, objects, or hand-written notes as live video

Rationale: A content camera (document camera or overhead camera positioned over a work surface) enables the teacher to share physical materials — textbook pages, student

work samples, 3D objects, live experiments — as a video feed to both in-room displays and remote participants. Option A describes a security camera. Option C describes presentation sharing (laptop output). Option D describes lecture capture only.

AVIXA Domain: Collaboration and UC Systems | ISTE Standard: 6 – Facilitator | *Bloom's Level: Understand*

Q84: Correct Answer: B — Including closed captioning display, assistive listening connectivity, and accessible UC platform features

Rationale: ISTE Standard 2 (Leader) emphasizes using technology to create equitable learning environments for all students, including those with disabilities. An accessible hybrid classroom AV design must include captioning (for deaf/hard-of-hearing students), assistive listening system connectivity (for hearing aid/cochlear implant users), and UC platform accessibility features (keyboard navigation, screen reader support). Options A, C, and D provide only partial accessibility solutions or exclude accessibility entirely.

AVIXA Domain: Collaboration and UC Systems | ISTE Standard: 2 – Leader | *Bloom's Level: Evaluate*

Q85: Correct Answer: B — 1-3 Mbps per session (up and down)

Rationale: Most video conferencing platforms (Zoom, Teams, Meet) use 1-2.5 Mbps for 720p HD video per participant pair (upstream and downstream combined). Ten simultaneous sessions at 2 Mbps each = 20 Mbps, well within 100 Mbps with headroom for other traffic. Option A (100 Kbps) is insufficient for HD video. Option C (10 Mbps per session) would require 100 Mbps just for video — leaving no headroom. Option D (50 Mbps) is the bandwidth needed for 4K uncompressed video, not compressed UC sessions.

AVIXA Domain: Collaboration and UC Systems | ISTE Standard: N/A | *Bloom's Level: Apply*

Domain 8: AV Design and Integration

Q86: Correct Answer: B — Identify functional requirements, workflows, user needs, and constraints to inform AV system design

Rationale: The needs assessment is the first step of any AV project, gathering requirements from stakeholders (teachers, administrators, students, IT staff) to define what the system must accomplish, who will use it, and within what budget and physical

constraints. Electrical panel analysis (A), brand selection (C), and as-built drawings (D) all come later in the project lifecycle.

AVIXA Domain: AV Design and Integration | ISTE Standard: 5 – Designer | *Bloom's Level: Understand*

Q87: Correct Answer: C — 1.5× screen height

Rationale: AVIXA/ANSI display standards specify that the minimum viewing distance (closest comfortable viewing distance) is 1.5 times the screen height (1.5H). Viewers closer than 1.5H may experience eye strain and will see individual pixels on lower-resolution displays. The maximum viewing distance is 6H (standard definition) or 8H (high definition). Options A (0.5H) and B (1H) would place viewers too close. Option D (3H) is between minimum and maximum, not the minimum.

AVIXA Domain: AV Design and Integration | ISTE Standard: N/A | *Bloom's Level: Remember*

Q88: Correct Answer: A — The screen bottom should be at least 1.2 meters (48 inches) above the finished floor

Rationale: The standard guideline (referenced in ANSI/AVIXA standards and widely adopted in AV design practice) sets the bottom edge of the screen at a minimum of 48 inches (approximately 1.2 meters) above the finished floor. This ensures seated audience members in the front rows can see over each other's heads. Options B and C tie screen height to the teacher's position, which does not account for audience sightlines. Option D would be too low.

AVIXA Domain: AV Design and Integration | ISTE Standard: N/A | *Bloom's Level: Remember*

Q89: Correct Answer: B — Long reverberation time, which reduces speech intelligibility

Rationale: Hard, parallel reflective surfaces (concrete, tile, glass) cause sound to bounce repeatedly before being absorbed, resulting in a long reverberation time (RT60). Long reverberation smears speech — words blur into each other, masking consonants and making the teacher difficult to understand, especially for students who are English language learners, have hearing differences, or are seated farther from the teacher. Option A is the opposite effect. Option C (no ambient noise) is physically impossible. Option D is a secondary consequence, not the primary acoustic problem.

AVIXA Domain: AV Design and Integration | ISTE Standard: 2 – Leader | *Bloom's Level: Understand*

Q90: Correct Answer: C — The AV floor plan / infrastructure drawing showing cable pathways and device locations

Rationale: The AV floor plan (often overlaid on the architectural floor plan) shows where each AV device is located, where cable pathways run through walls, ceilings, and conduit, and identifies connection points. This document is essential for coordination with other trades (electrical, structural, mechanical) during construction. Equipment rack elevations (A) show how equipment is arranged in a rack. Signal flow diagrams (B) show signal routing, not physical pathways. Specification sheets (D) describe individual equipment performance.

AVIXA Domain: AV Design and Integration | ISTE Standard: N/A | Bloom's Level: Apply

Q91: Correct Answer: B — 15 to 48 inches AFF

Rationale: ADA Standards for Accessible Design (Section 4.2) specify that operable elements (including touch panels) must be within reach ranges: forward reach 15–48 inches AFF (without obstruction), and side reach with obstruction may vary. The 15–48 inch range (approximately 38–122 cm) is the commonly cited standard for AV control interface mounting height to ensure wheelchair users can reach and operate controls. Options A, C, and D do not accurately represent the ADA-specified range.

AVIXA Domain: AV Design and Integration | ISTE Standard: 2 – Leader | Bloom's Level: Remember

Q92: Correct Answer: B — Functional design / user workflow accommodation

Rationale: Incorporating a document camera above a demonstration bench to support a specific teacher workflow (showing experiments to the class) is a functional design decision made during the design phase when user workflows are analyzed and translated into system requirements and equipment specifications. Commissioning (A) occurs after installation. Preventive maintenance planning (C) occurs post-installation. Network infrastructure provisioning (D) relates to IT planning, not the camera's functional placement.

AVIXA Domain: AV Design and Integration | ISTE Standard: 5 – Designer | Bloom's Level: Apply

Q93: Correct Answer: B — 1.75 inches (44.45 mm)

Rationale: One rack unit (1U) is defined as 1.75 inches (44.45 mm) in height, derived from the EIA-310 standard for equipment racks. This standardization allows AV equipment from different manufacturers to be mounted in standard 19-inch-wide racks. A 4U device is $4 \times 1.75 = 7$ inches tall. Knowing rack unit dimensions is essential for rack layout planning in AV design.

AVIXA Domain: AV Design and Integration | ISTE Standard: N/A | Bloom's Level: Remember

Q94: Correct Answer: B — Verifying, testing, adjusting, and documenting that installed AV systems meet design specifications before handover

Rationale: Commissioning is the systematic process of verifying that all installed components function correctly, that the system as a whole meets the design specifications, and that all performance parameters (audio levels, video signal integrity, control system responses) are measured, adjusted, and documented. Physical installation (A) precedes commissioning. Needs assessment (C) precedes design. Procurement (D) precedes installation.

AVIXA Domain: AV Design and Integration | ISTE Standard: N/A | Bloom's Level: Remember

Q95: Correct Answer: C — Open standards, modular architecture, and adequate conduit/cable capacity for future upgrades

Rationale: Scalable AV design uses open standards (so future equipment from any manufacturer can integrate), modular architecture (so components can be swapped without redesigning the entire system), and oversized conduit runs and cable capacity (so new cables can be pulled in the future without costly construction). Option A (cheapest equipment) prioritizes short-term cost over long-term value. Option B (proprietary systems) creates vendor lock-in. Option D (maximum resolution displays) addresses content quality, not scalability.

AVIXA Domain: AV Design and Integration | ISTE Standard: 5 – Designer | Bloom's Level: Evaluate

Domain 9: Maintenance and Troubleshooting

Q96: Correct Answer: B — Replace the lamp proactively before it fails, per the manufacturer's maintenance schedule

Rationale: Projector lamps should be replaced proactively based on the manufacturer's recommended hour rating (often 2,000–5,000 hours), not run to failure. A failed lamp during class causes unplanned downtime, potentially during a critical lesson or presentation. Spare lamps should be kept on hand. Waiting for failure (A) is reactive maintenance. Cleaning a lamp (C) is dangerous and ineffective — lamp envelopes should not be touched. Increasing fan speed (D) does not prevent lamp failure.

AVIXA Domain: Maintenance and Troubleshooting | ISTE Standard: N/A | *Bloom's Level: Apply*

Q97: Correct Answer: B — Gathering information — identifying and defining the problem clearly

Rationale: A structured troubleshooting methodology (consistent with CompTIA, AVIXA, and industry best practices) begins with information gathering: what exactly is the problem, when did it start, what changed recently, what error messages appear, and is it reproducible? This step prevents wasted effort on the wrong problem. Replacing expensive components first (A) is wasteful and unprofessional. Rebooting all devices (C) may help some issues but is not the first step. Calling the manufacturer (D) is appropriate only after own-analysis is exhausted.

AVIXA Domain: Maintenance and Troubleshooting | ISTE Standard: N/A | *Bloom's Level: Apply*

Q98: Correct Answer: B — Test the video signal path by substituting a known-good cable and/or source to isolate the fault

Rationale: After confirming the display is on and the input is correct, the systematic next step is component substitution to isolate the fault: try a known-good cable; try a known-good source device. This narrows the problem to a specific cable, port, or device. Replacing the display immediately (A) is premature and costly without isolating the fault. Control system firmware (C) and the electrical panel (D) are unrelated to a video-only failure in this scenario.

AVIXA Domain: Maintenance and Troubleshooting | ISTE Standard: N/A | *Bloom's Level: Apply*

Q99: Correct Answer: B — Cleaning projector filters, checking lamp hours, inspecting cable connections, testing control system, and updating firmware

Rationale: Preventive maintenance (PM) for AV systems involves a scheduled set of tasks that preserve system health and prevent failures. Core tasks include: cleaning projector air filters (prevents overheating), checking and logging lamp hours (enables proactive replacement), inspecting all cable connections (prevents intermittent signal issues), testing control system macros (ensures programming integrity), and applying firmware updates (patches vulnerabilities and bugs). Options A, C, and D describe either excessive actions (replacing all cables annually) or reactive behaviors (replacing batteries only when depleted).

AVIXA Domain: Maintenance and Troubleshooting | ISTE Standard: N/A | *Bloom's Level: Apply*

Q100: Correct Answer: B — Creates institutional knowledge base and records for warranty and maintenance contracts

Rationale: Thorough troubleshooting documentation serves multiple critical functions: it creates a searchable record that helps future technicians recognize and resolve recurring issues faster; it provides evidence for warranty claims; it satisfies requirements of service-level agreements and maintenance contracts; and it supports budget justification for equipment replacement. Option A is partially valid but too narrow. Option C is not an AVIXA AVT certification requirement. Option D is false — documentation supplements but does not replace preventive maintenance.

AVIXA Domain: Maintenance and Troubleshooting | ISTE Standard: N/A | *Bloom's Level: Evaluate*

Quick-Check Bank Answer Key (Q101-Q120)

Q101 (Unit 1): Correct Answer: C — AV industry value chain

Rationale: The AV industry value chain encompasses the full ecosystem of manufacturers, distributors, integrators, consultants, and service providers whose combined contributions deliver AV solutions to end users. "Signal chain" (A) refers to the path of an AV signal through equipment. "AV infrastructure" (B) refers to physical installation components. "Protocol stack" (D) is a networking term.

AVIXA Domain: AV Industry Overview & Fundamentals | ISTE Standard: N/A | *Bloom's Level: Remember*

Q102 (Unit 1): Correct Answer: B — Ensuring AV and IT systems are accessible to people with disabilities

Rationale: Section 508 of the Rehabilitation Act requires that federal agencies (and entities receiving federal funding, including most public schools) ensure their electronic and information technology — including AV systems — is accessible to people with disabilities. This covers assistive technology compatibility, captioning, accessible interfaces, and ADA-compliant control placement.

AVIXA Domain: AV Industry Overview & Fundamentals | ISTE Standard: 2 – Leader | *Bloom's Level: Remember*

Q103 (Unit 1): Correct Answer: B — RF interference or multipath dropouts in the wireless microphone's radio frequency transmission

Rationale: Wireless microphone dropouts occur when the RF signal path between the transmitter (in the microphone) and receiver is disrupted by RF interference from competing devices, or by multipath cancellation (reflected signals arriving out of phase and canceling each other). Large gymnasium metal structures exacerbate multipath. Condenser microphone type (A) does not cause dropouts. Phantom power on a receiver (C) is not an applicable concept. Impedance mismatch (D) affects wired connections, not wireless transmission.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | *Bloom's Level: Apply*

Q104 (Unit 1): Correct Answer: B — The signals cancel each other out, causing reduced bass and a thin, hollow sound

Rationale: Phase cancellation occurs when two identical signals are combined with one inverted (180° out of phase). At points of complete cancellation, the result is silence; partial cancellation produces frequency-dependent comb filtering — characterized by a hollow, thin, phasey sound with notable bass reduction. Option A describes in-phase summing. Option C describes only partial effects. Option D is a false claim — phase affects time relationships, not impedance values.

AVIXA Domain: Audio Systems | ISTE Standard: N/A | *Bloom's Level: Understand*

Q105 (Unit 2): Correct Answer: B — 1920 × 1080 (16:9)

Rationale: A 16:9 screen will display 16:9 content without bars or distortion. 1920×1080 (Full HD) has a 16:9 aspect ratio and is appropriate for standard HD projection in auditoriums. Options A and C (both 4:3) would produce letterbox bars on a 16:9 screen.

Option D (2560×1080 ultra-wide) is a 21:9 aspect ratio that would produce pillar bars on a 16:9 screen.

AVIXA Domain: Video and Display Systems | ISTE Standard: N/A | *Bloom's Level: Apply*

Q106 (Unit 2): Correct Answer: B — Image burn-in (phosphor or pixel retention)

Rationale: Image burn-in occurs when static content (such as a fixed menu, logo, or classroom display layout) is shown for extended periods on certain display technologies (OLED, plasma, and to a lesser extent some LCD panels). The static elements become permanently "burned" into the display material, remaining visible even when the display content changes. This is a known risk for displays used for digital signage or static classroom content over many years.

AVIXA Domain: Video and Display Systems | ISTE Standard: N/A | *Bloom's Level: Analyze*

Q107 (Unit 2): Correct Answer: B — The signal flow chain: source → switch → amplify → output

Rationale: Source, switcher, amplifier, and display are the four classic stages of the AV signal flow model: a signal originates at a source (laptop, camera), is selected/routed by a switcher, amplified or processed, and delivered to an output (display, speaker). This model is fundamental to both understanding and troubleshooting AV systems. Options A, C, and D are unrelated frameworks.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | *Bloom's Level: Understand*

Q108 (Unit 2): Correct Answer: B — RF interference or physical obstructions in the 5 GHz transmission path

Rationale: Wireless HDMI extenders typically operate on 5 GHz or 60 GHz (WiGig) bands. Physical obstructions (people walking through the path, furniture, walls) and interference from other 5 GHz Wi-Fi devices or microwave-band equipment can cause signal degradation and video freezing. Display resolution (A) and HDCP version (C) would cause persistent failure, not intermittent freezing. Screen reflectivity (D) is a video quality factor, not a transmission issue.

AVIXA Domain: Signal Flow and Connectivity | ISTE Standard: N/A | *Bloom's Level: Analyze*

Q109 (Unit 3): Correct Answer: B — Uncompressed 4K video requires approximately 12 Gbps per stream, far exceeding a 1 Gbps switch

Rationale: Uncompressed 4K/30Hz 4:4:4 video requires approximately 6–12 Gbps depending on color depth and frame rate. A 1 Gbps switch cannot handle even a single uncompressed 4K stream. The solution is either to use a 10 Gbps (or 25 Gbps) network infrastructure for AVoIP or to deploy a compressed 4K AVoIP system (e.g., JPEG 2000 at 1 Gbps). Options A, C, and D do not correctly describe the underlying bandwidth constraint.

AVIXA Domain: AV Networking | ISTE Standard: N/A | Bloom's Level: Analyze

Q110 (Unit 3): Correct Answer: B — APIPA address indicating DHCP or network connectivity failure

Rationale: The 169.254.0.0/16 address range is reserved for APIPA (Automatic Private IP Addressing), also called link-local addressing. When a device cannot reach a DHCP server, it auto-assigns a 169.254.x.x address, allowing local-link communication only. This is a clear diagnostic indicator that the device cannot get a DHCP lease — check that the DHCP server is reachable, the device is on the correct VLAN, and the network cable or Wi-Fi association is functional.

AVIXA Domain: AV Networking | ISTE Standard: N/A | Bloom's Level: Analyze

Q111 (Unit 3): Correct Answer: B — Control processor communication settings incorrect or control program not uploaded

Rationale: If the touch panel displays the correct user interface (meaning the touch panel itself is communicating with the control processor), but no commands are executing, the most likely causes are: the control processor's IP address or serial (RS-232) settings don't match the AV devices' expected settings, or the control program was not compiled and uploaded to the processor. The HDMI cable (A), ambient temperature (C), and firmware versions (D) would not cause a scenario where the GUI appears correctly but commands fail.

AVIXA Domain: Control Systems | ISTE Standard: N/A | Bloom's Level: Analyze

Q112 (Unit 3): Correct Answer: B — Relay trigger not sent, relay wiring incorrect, or relay contact damaged

Rationale: A motorized screen typically uses a dry-contact relay to close the screen motor's "down" circuit. Troubleshooting should verify: (1) the control program is sending the relay-trigger command during the "Projector On" macro, (2) the relay output on the control processor is wired correctly to the screen motor's input terminals, and (3) the relay contact itself is not damaged (use a multimeter to test continuity when the relay should be triggered). The projector lamp (A), HDMI aspect ratio (C), and network cable (D) are unrelated to relay/screen operation.

AVIXA Domain: Control Systems | ISTE Standard: N/A | *Bloom's Level: Analyze*

Q113 (Unit 4): Correct Answer: B — Positioning the camera so remote participants see clearly while in-room students can make natural eye contact with remote participants on the display

Rationale: The camera placement problem in hybrid classrooms involves balancing two competing needs: the camera must capture the teacher and in-room students clearly for remote viewers, while the display showing remote participants must be positioned so in-room students can make near-eye-contact (the camera ideally at or near the display). Ideal placement puts the camera adjacent to or integrated with the far-end display. Options A, C, and D address different camera characteristics, not the placement challenge.

AVIXA Domain: Collaboration and UC Systems | ISTE Standard: 2 - Leader | *Bloom's Level: Understand*

Q114 (Unit 4): Correct Answer: B — DSP algorithms (and AI-based models) that distinguish speech from non-speech sounds and attenuate background noise in real time

Rationale: Modern UC noise cancellation uses digital signal processing — and increasingly AI/machine learning models trained on vast audio datasets — to classify audio as speech or noise in real time, attenuating non-speech signals (keyboard clicks, HVAC, ambient conversation) before transmission. Option A (physical barrier) describes acoustic isolation. Option C (active noise control speaker) is a different technology used in headphones. Option D (reducing gain) would reduce both speech and noise.

AVIXA Domain: Collaboration and UC Systems | ISTE Standard: N/A | *Bloom's Level: Understand*

Q115 (Unit 4): Correct Answer: B — RT60 between 0.4 and 0.6 seconds

Rationale: RT60 (reverberation time) is the time for sound to decay by 60 dB after the source stops. For classrooms, ANSI/ASA S12.60 (the standard for classroom acoustics) recommends $RT60 \leq 0.6$ seconds for smaller classrooms and ≤ 0.7 seconds for larger ones. The 0.4–0.6 second range is optimal — short enough for speech clarity, long enough to avoid sounding overly "dead." Less than 0.2 seconds (A) is too dry. 1.5–2.0 seconds (C) suits concert halls. Greater than 3.0 seconds (D) is typical of reverberant spaces like gymnasiums.

AVIXA Domain: AV Design and Integration | ISTE Standard: N/A | Bloom's Level: Remember

Q116 (Unit 4): Correct Answer: B — 2 meters

Rationale: If the maximum viewing distance is 6H (6 times screen height), and the room is 12 meters deep, then: $12 \text{ m} \div 6 = 2$ meters minimum screen height. A screen height of 2 meters would allow the viewer at the back row (12 m away) to still be within the 6H maximum viewing distance. A 1-meter screen (A) would only serve viewers within 6 meters. A 3-meter screen (C) is larger than necessary. A 6-meter screen (D) would be enormous for a standard classroom.

AVIXA Domain: AV Design and Integration | ISTE Standard: N/A | Bloom's Level: Apply

Q117 (Unit 5): Correct Answer: B — As-built drawings, equipment manuals, warranty documentation, control system source code, cable schedules, and training record

Rationale: A professional AV project handover package includes: as-built drawings (reflecting what was actually installed), equipment manuals, warranty cards/registrations, control system source code (critical for future modifications), cable schedules, and a record of user training. Purchase receipts (A) are financial documents, not operational documentation. The original design proposal (C) reflects the plan, not the final installation. A single unlabeled diagram (D) is insufficient for ongoing support.

AVIXA Domain: AV Design and Integration | ISTE Standard: N/A | Bloom's Level: Apply

Q118 (Unit 5): Correct Answer: B — Check whether the laptop is outputting video to the HDMI port (display output settings)

Rationale: After ruling out the cable as the cause (already tested a different cable), the next logical step is to verify the source — specifically whether the laptop is configured to

send a signal to its HDMI output. Many laptops default to "laptop screen only" mode; the user must press the display-switch key combination (Fn + display key, or the Windows key + P on Windows) to switch to "Duplicate," "Extend," or "Second Screen Only" mode. Display audio muting (A) does not affect video output. Lamp hours (C) and network IP (D) are unrelated to this video-specific symptom.

AVIXA Domain: Maintenance and Troubleshooting | ISTE Standard: N/A | *Bloom's Level: Apply*

Q119 (Unit 5): Correct Answer: B — Cleaning projector filters, checking lamp hours, inspecting cable connections, and testing wireless microphone battery life

Rationale: Semi-annual (twice-yearly) preventive maintenance for school AV systems typically includes: cleaning projector air filters (prevents thermal shutdown), checking and logging lamp hours against replacement thresholds, visually inspecting cable connections for seating and wear, and testing/replacing wireless microphone batteries proactively. Replacing all displays (A) and all wireless transmitters (C) annually would be extremely cost-prohibitive and unnecessary. Reprogramming all touch panels (D) is not a maintenance task unless there are identified issues.

AVIXA Domain: Maintenance and Troubleshooting | ISTE Standard: N/A | *Bloom's Level: Apply*

Q120 (Unit 5): Correct Answer: B — Rotate the image 180 degrees using the document camera's built-in rotation setting

Rationale: Document cameras include a built-in image rotation feature specifically to handle situations where the camera head is oriented differently from the intended viewing direction. A simple 180-degree rotation correction in the camera's settings or physical controls resolves the upside-down image instantly without hardware replacement (A), programming changes (C), or keystone adjustment (D, which corrects geometric distortion, not rotation).

AVIXA Domain: Maintenance and Troubleshooting | ISTE Standard: N/A | *Bloom's Level: Apply*

Section 4: Domain Performance Analysis Worksheet

Use this worksheet after completing and scoring the 100-question practice exam to identify your strengths and areas for additional study. Transfer your scores by domain below, then use the analysis to build a targeted study plan.

How to Complete This Worksheet

1. Go through your completed exam, sorted by domain (use the domain tags on each question).
2. Count how many questions in each domain you answered correctly.
3. Divide your domain score by the total questions in that domain and multiply by 100 to get your percentage.
4. Mark each domain as a Strength ($\geq 75\%$) or Needs Work ($< 75\%$).
5. Use the recommended resources column to focus your review.

Self-Scoring Table: 100-Question Practice Exam

Domain	Questions in Exam	My Correct Answers	Percentage (%)	Strength / Needs Work
AV Industry Overview & Fundamentals	10			
Audio Systems	15			
Video and Display Systems	15			
Signal Flow and Connectivity	15			
AV Networking	10			
Control	10			

Domain	Questions in Exam	My Correct Answers	Percentage (%)	Strength / Needs Work
Systems				
Collaboration and UC Systems	10			
AV Design and Integration	10			
Maintenance and Troubleshooting	5			
TOTAL EXAM SCORE	100			

Recommended Study Resources by Domain

Domain	Textbook Chapter(s)	Recommended Focus Topics	Additional Resources
AV Industry Overview & Fundamentals	Ch. 1	AV ecosystem, digital vs. analog, UDL, ISTE Standards, AV careers, ADA/508, needs assessment	AVIXA.org Industry Fundamentals; ISTE Standards website
Audio Systems	Ch. 2	Decibels, microphone types, polar patterns, phantom power, impedance, frequency response, AEC, DSP, assistive listening, speaker placement	AVIXA CTS Study Guide Ch. 2; AES.org fundamentals
Video and Display Systems	Ch. 3	Lumens, throw ratio, native resolution, 4K/HD standards, contrast ratio, aspect ratio, display types, WCAG, viewing distances, keystone, digital signage	AVIXA display standards; manufacturer spec sheets; WCAG 2.1 documentation
Signal Flow and Connectivity	Ch. 4	HDMI, DisplayPort, HDBaseT, USB-C, XLR, balanced/unbalanced audio, EDID,	HDBaseT Alliance technical resources; HDMI.org specification overview

Domain	Textbook Chapter(s)	Recommended Focus Topics	Additional Resources
		HDCP, DA, matrix switchers, cable labeling, ground loops	
AV Networking	Ch. 5	IP addressing, DHCP, VLAN, QoS, multicast, IGMP, bandwidth calculations, managed vs. unmanaged switches, network security	AVIXA AV Networking Fundamentals course; CompTIA Network+ study resources
Control Systems	Ch. 6	Control processors, RS-232, IR control, touch panel UI design, macros, occupancy sensors, APIs, cybersecurity, relays, serial communication parameters	AVIXA CTS Study Guide Ch. 6; control system manufacturer documentation
Collaboration and UC Systems	Ch. 7	Hybrid equity, AEC, codecs, PTZ cameras, UC platforms, SIP/H.323, bandwidth for video conferencing, noise cancellation, content cameras, accessibility	AVIXA Collaboration Systems resources; UC platform knowledge bases
AV Design and Integration	Ch. 8-9	Needs assessment, sightlines, screen height, viewing distances, room acoustics/RT60, ADA compliance, rack units, commissioning, system documentation, scalable design	ANSI/AVIXA 1M-2021 display standard; ANSI/ASA S12.60 classroom acoustics; ADA.gov
Maintenance and Troubleshooting	Ch. 10	Preventive maintenance schedules, lamp replacement, filter cleaning, structured troubleshooting methodology, component	AVIXA CTS Study Guide Ch. 10; manufacturer maintenance guides

Domain	Textbook Chapter(s)	Recommended Focus Topics	Additional Resources
		substitution, documentation	

Study Planning Template

Use your domain analysis to build a personalized study schedule. Prioritize domains marked "Needs Work," spending the most time on your lowest-scoring areas.

Domains marked "Strength" need only light review before the exam.

Study Week	Primary Domain Focus	Textbook Chapters	Study Activities	Completed?
Week 1	Top "Needs Work" Domain #1: _____		Read chapter, review rationales for missed questions, online resources	☐
Week 2	Top "Needs Work" Domain #2: _____		Read chapter, review rationales for missed questions, flashcards	☐
Week 3	Top "Needs Work" Domain #3: _____		Read chapter, lab/hands-on practice if available	☐
Week 4	All remaining domains + comprehensive review	All Ch. 1-10	Full practice exam retake; review all rationales	☐
Week 5	Final review: all domains	All Ch. 1-10	Strength domain quick-checks; weak domain deep dive; schedule official exam	☐

My Weakest Domains (Needs Work) — Priority Study List

List your lowest-scoring domains here to keep them visible during study sessions:

1. _____

2. _____
3. _____
4. _____

My Strongest Domains — Light Review Needed

1. _____
2. _____

Target Exam Date and Score Goal

Official AVIXA AV Technologist Exam Date: _____

My Practice Exam Score: _____ / 100 | My Target Improvement: + _____ points

Date to Retake This Practice Exam: _____

Section 5: Instructor Notes for Assessment Use

This section provides guidance for educators and AV program instructors on how to effectively deploy this assessment bank in a variety of instructional contexts. The strategies below reflect both best practices in educational assessment design and the professional development context of AV training for educators.

Administering the 100-Question Exam as a Timed Practice

Recommended Time Allowance

90 minutes is the recommended time for the full 100-question practice exam under simulated testing conditions. This mirrors the typical time allocation of the official AVIXA AV Technologist examination. At 54 seconds per question, students who have studied the material should be able to complete all questions within this timeframe while still reviewing flagged items.

Timing Breakdown Recommendation

- **0:00-75:00** — Work through all 100 questions; mark uncertain answers
- **75:00-88:00** — Return to marked questions for a second pass
- **88:00-90:00** — Final check; do not change answers unless clearly incorrect

Physical Administration

- Print questions on single-sided paper and provide a separate answer sheet (grid format, A-D for Q1-Q100).
- For paper administration, collect all question packets to prevent sharing between sections.
- For digital administration via LMS, refer to the QTI import guidance below.
- Students should not have access to textbooks, notes, or devices during the timed exam simulation.
- After the exam, students should receive both the question packet and a scored answer key to review with rationales.

Using This Bank as a Pre-Test / Post-Test

Pre-test/post-test design is the most powerful way to demonstrate student growth and identify knowledge gaps at course entry. This assessment bank supports this model in two ways:

Option A: Full-Bank Pre/Post

Administer the full 100-question exam on Day 1 of the course (pre-test) and again in the final week (post-test). Score by domain both times and compare the Domain Analysis Worksheet results. Students and instructors will see measurable growth in each domain, providing powerful evidence of learning and informing any final instructional adjustments before the official exam.

Option B: Domain-Sampled Pre/Post

Select 10–15 questions from each of the highest-weight domains (Audio, Video, Signal Flow) for a 30–45 question pre-test administered in the first week. Administer the same questions as a post-test in the final week. This is less fatiguing for students at course entry and still provides meaningful pre/post comparison data.

Pre-Test Administration Note

: Inform students that the pre-test score will **not** affect their course grade — it is diagnostic only. This reduces test anxiety and produces more authentic baseline data.

Selecting Question Subsets by Domain for Chapter Quizzes

Each question in this bank is tagged with an **AVIXA Domain** and **Chapter Reference**, making it straightforward to extract questions for chapter-level quizzes. Use the table below as a quick reference for quiz construction.

Quiz Purpose	Recommended Question Count	Questions to Draw From	Suggested Time
Ch. 1 Exit Quiz (AV Fundamentals)	5–8 questions	Q1–Q10, Q101–Q102	8–12 minutes
Ch. 2 Quiz (Audio Systems)	8–10 questions	Q11–Q25, Q103–Q104	12–15 minutes
Ch. 3 Quiz (Video & Display)	8–10 questions	Q26–Q40, Q105–Q106	12–15 minutes
Ch. 4 Quiz (Signal Flow)	8–10 questions	Q41–Q55, Q107–Q108	12–15 minutes
Ch. 5 Quiz (AV Networking)	5–8 questions	Q56–Q65, Q109–Q110	8–12 minutes
Ch. 6 Quiz (Control Systems)	5–8 questions	Q66–Q75, Q111–Q112	8–12 minutes
Ch. 7 Quiz (Collaboration & UC)	5–8 questions	Q76–Q85, Q113–Q114	8–12 minutes

Quiz Purpose	Recommended Question Count	Questions to Draw From	Suggested Time
Ch. 8-9 Quiz (Design & Integration)	5-8 questions	Q86-Q95, Q115-Q116	8-12 minutes
Ch. 10 Quiz (Maintenance & Troubleshooting)	4-5 questions	Q96-Q100, Q117-Q120	6-8 minutes
Mid-Course Exam (Ch. 1-5)	30-40 questions	Q1-Q65 (sample by domain weight)	40-50 minutes
Final Comprehensive Exam	50-100 questions	Full bank, all domains	60-90 minutes

Bloom's Level Selection

When constructing quizzes, aim to mirror the exam's Bloom's distribution: approximately 40% recall/remember questions, 40% application questions, and 20% analysis/evaluation questions. The Bloom's Level tag in the Answer Key (Section 3) allows precise selection. For lower-stakes weekly quizzes, a higher proportion of recall questions is appropriate. For summative assessments, prioritize application and analysis questions.

LMS Import Guidance

This question bank is available as a **QTI 2.1-compatible XML package** through the online resource portal associated with this course. QTI (Question and Test Interoperability) is an IMS Global standard for exchanging assessment content between learning management systems.

Supported LMS Platforms

LMS Platform	QTI Version Supported	Import Path	Notes
Canvas (Instructure)	QTI 1.2 / 2.1	Settings → Import Course Content → QTI .zip file	Import as a Question Bank; questions import to "Unfiled Questions" by default
Schoology (PowerSchool)	QTI 2.1	Courses → Resources → Add →	Verify question type mapping after

LMS Platform	QTI Version Supported	Import Path	Notes
		Import from QTI	import
Moodle	Moodle XML / QTI 2.1	Question Bank → Import → Select QTI 2.0 format	Use the Moodle-specific export from the resource portal for best compatibility
Blackboard Learn	QTI 1.2 / Blackboard Pool	Course Tools → Tests, Surveys, and Pools → Pool Manager → Import	Use Blackboard-format export if available; verify after import
Google Forms	N/A (manual entry)	Manual or via third-party QTI-to-Forms converter	For Google Forms, use the formatted question list in this PDF and a QTI converter tool

After Import — Verify These Settings

- Confirm that correct answers are properly marked in the LMS quiz editor.
- Add question rationales to the LMS "General Feedback" field for each question so students see rationales after submission.
- Set the quiz to display questions in random order to prevent answer-sharing between students in different sections.
- Enable "one question at a time" display to prevent students from jumping ahead and spending less time on early questions.
- Configure the quiz to show correct answers and feedback only after the quiz has closed (not immediately), to preserve the bank's integrity across course sections.

Accommodations Guidance

This section provides guidance for providing assessment accommodations consistent with ADA, Section 504, and IDEA requirements. Instructors should consult their institution's disability services office for official accommodation plans and documentation requirements.

Extended Time

The most common accommodation for timed assessments is extended time — typically 1.5× (time and a half) or 2× the standard time allocation. Apply these multipliers as follows:

Accommodation	Standard Time (100 Qs)	1.5× Extended Time	2× Extended Time
Full 100-Question Exam	90 minutes	135 minutes	180 minutes
Unit Quick-Check (4 Qs)	8–10 minutes	12–15 minutes	16–20 minutes
Chapter Quiz (10 Qs)	15 minutes	22–23 minutes	30 minutes

When administering with extended time in a digital LMS environment, simply adjust the quiz time limit in the LMS settings for the specific student's quiz attempt. Most LMS platforms support per-student time extensions without creating a separate quiz version.

Read-Aloud Accommodations

For students who receive read-aloud accommodations:

- **Human reader:** The reader should read the question stem and all four answer options aloud verbatim, without inflection changes that might hint at the correct answer. The reader should not re-read or clarify questions unless the student requests it.
- **Text-to-speech (TTS):** Most LMS platforms support browser-based or built-in TTS. Instruct the student to enable TTS in their browser or screen reader before beginning the quiz. The question text in this bank has been written to be TTS-compatible (no special characters, symbols spelled out).

- **Screen reader:** Ensure the LMS quiz interface is ARIA-compliant and keyboard-navigable. Questions in this bank use plain text without tables or complex formatting in the question stems, supporting screen reader compatibility.

Alternative Formats

- **Large Print:** Increase the body font size in the HTML document to 14pt or 16pt and reprint. The document's responsive layout will accommodate this adjustment.
- **Braille:** Contact your institution's disability services office to arrange Braille transcription. Provide this source document in plain-text format for the transcription service.
- **Reduced-distraction environment:** Arrange for students with this accommodation to complete the exam in a separate, quiet location with standard or extended time as indicated on their accommodation plan.
- **Oral response:** For students with writing disabilities, allow verbal responses recorded by a proctor or digitally recorded and transcribed. The multiple-choice format (A/B/C/D) simplifies oral response administration.

Assistive Technology in AV Labs

For any hands-on assessment components conducted alongside this written exam, ensure that the AV lab environment itself is accessible:

- AV control panels at wheelchair-accessible height (15–48 inches AFF).
- Equipment labels available in large print or digital format for students with visual impairments.
- Clear floor pathways with no cable trip hazards for students using mobility aids.
- Assistive listening available in the lab space for students who are deaf or hard of hearing.

✓ Instructor Reflection: Assessment as a Learning Tool

Research consistently shows that retrieving information from memory (the "testing effect" or "retrieval practice effect") is one of the most powerful strategies for long-term retention. Encourage students to treat this practice exam — and especially the answer key rationales — as a primary study tool, not just a performance measure. Students who carefully read every rationale, including for questions they answered correctly, demonstrate significantly better outcomes on high-stakes assessments. Consider building a structured post-exam rationale review session into your course schedule.

Instructor Quick-Reference: Question Distribution by Bloom's Level

Bloom's Level	Approximate % of 100-Q Exam	Approximate Question Count	Representative Questions
Remember (Recall)	~40%	~40 questions	Q1, Q3, Q12, Q15, Q16, Q18, Q19, Q27, Q29, Q30, Q34, Q42, Q50, Q57, Q60, Q66, Q67, Q69, Q78, Q87, Q88, Q93, Q94
Understand / Apply	~40%	~40 questions	Q2, Q4, Q5, Q6, Q7, Q8, Q11, Q13, Q20, Q21, Q22, Q24, Q26, Q28, Q37, Q44, Q45, Q51, Q53, Q58, Q59, Q61, Q62, Q63, Q68, Q71, Q72, Q73, Q76, Q79, Q81, Q82, Q85, Q86, Q90, Q92, Q96, Q97, Q98, Q99
Analyze / Evaluate	~20%	~20 questions	Q10, Q17, Q25, Q31, Q36, Q43, Q47, Q49, Q55, Q64, Q65, Q70, Q74, Q84, Q89, Q95, Q100

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AV Fundamentals for Educators: Practice Exam and Assessment Bank

ISTE × AVIXA Educational Resource Series | May 2026

120 Questions | 9 AVIXA Domains | Chapters 1-10

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